

PLAN SET

Workbench Plans

A 69" x 96" island at SawStop height: outfeed, router lift station, flip-top miter saw, downdraft clamping station, laser well, built-in ducting and power.

Owner: 1688ross · Generated October 01, 2026 from tools/model.py · All dimensions in inches · github.com/1688ross/workbench-plans

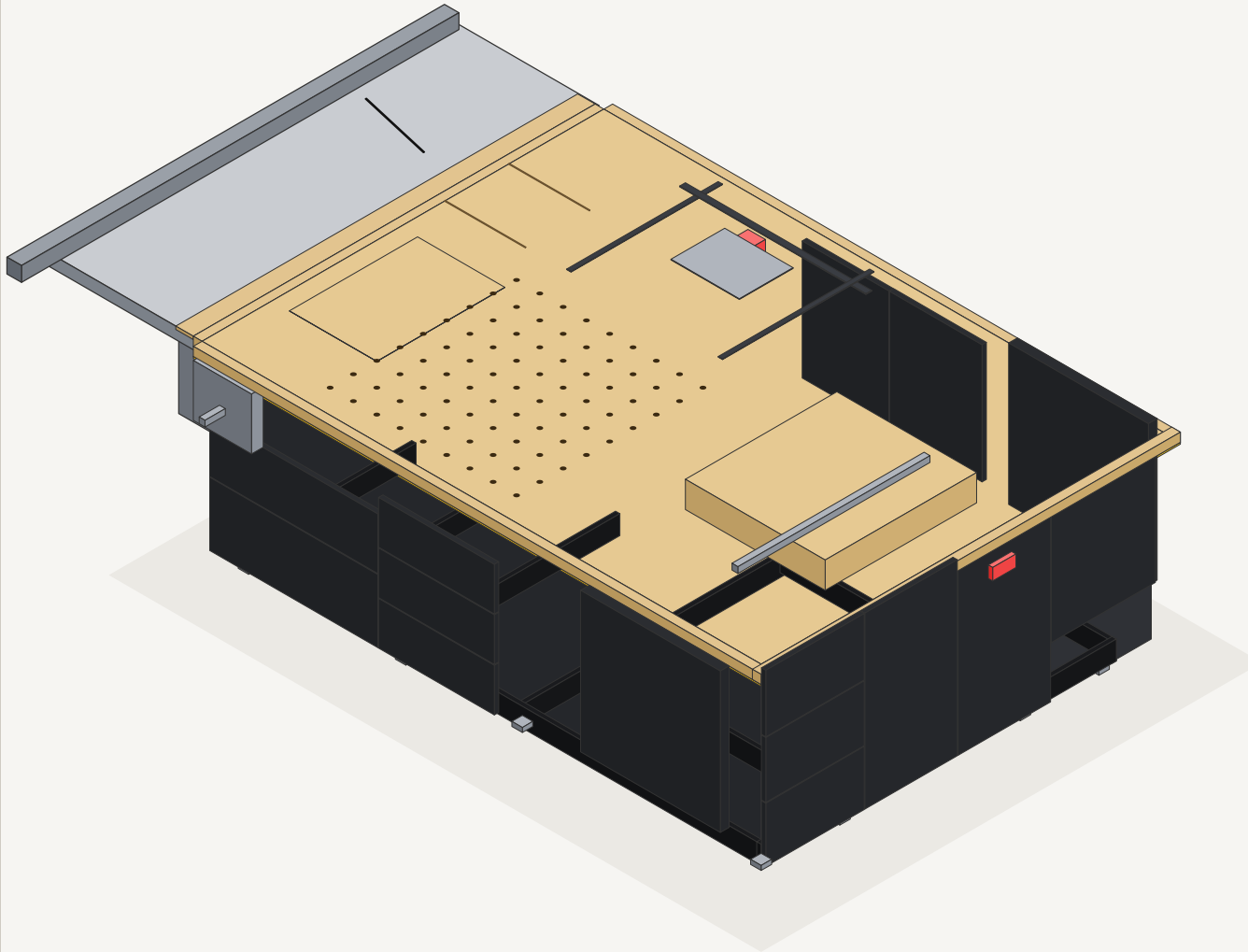
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Measure before you cut: the seven items on the VERIFY list in Design decisions. Every number in this set comes from one model file, so a changed measurement regenerates every drawing and the cut list together.

FINAL LOOK - from the south-west, everything stowed

Charcoal Baltic-birch fronts with routed finger pulls, maple-edged 1-1/2" top at 3/4". Miter saw hangs upside down under its flush box; only the router plate, T-track, dog field and vise show.

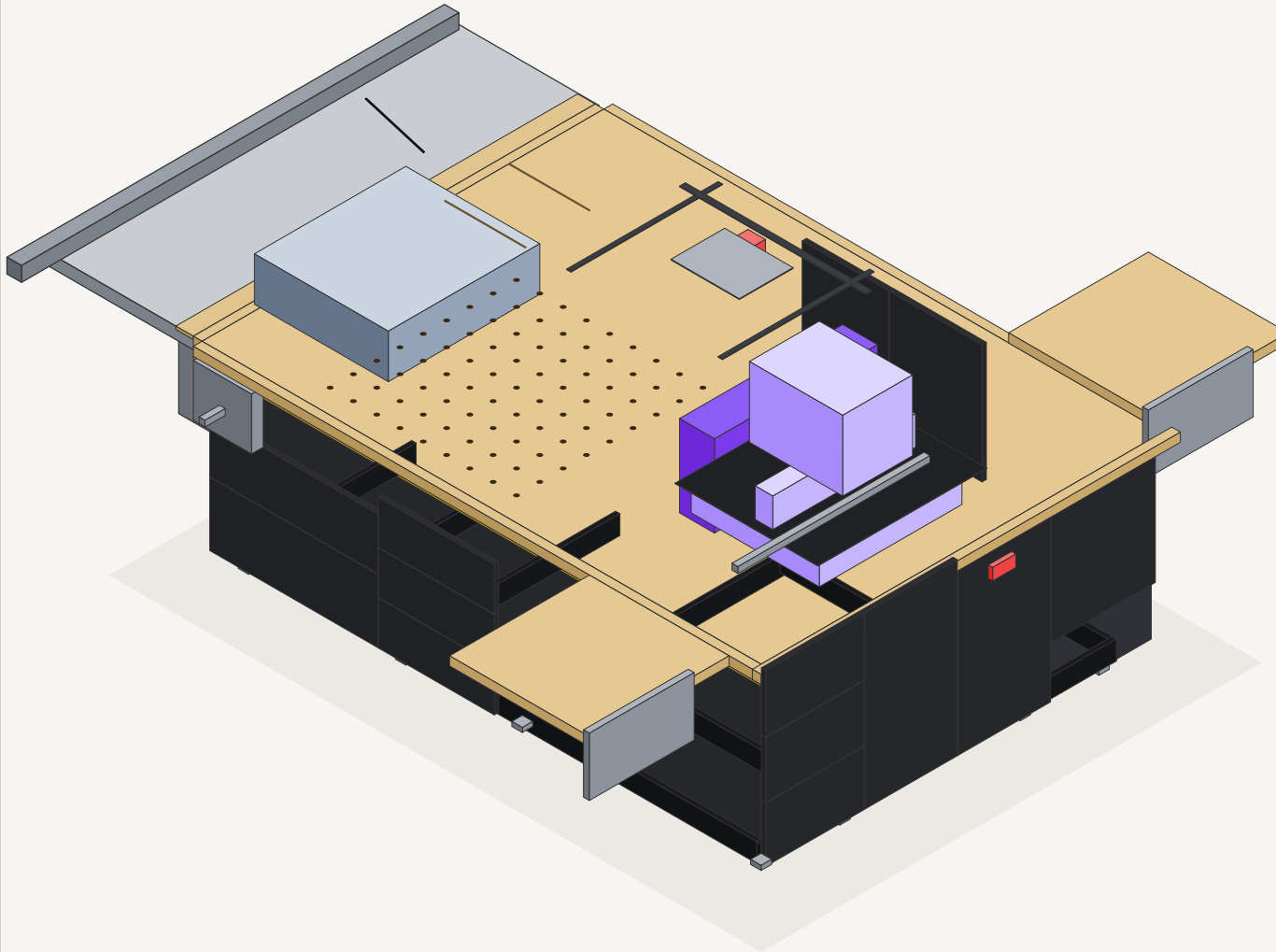


West face (left): face vise, dog field, hand-tool drawers, laser tray. South face (right): long drawers, two flip-bay doors with the miter paddle, clamp-rack pull-out. Drop-leaf wings hang flat against both ends. LED strip under the top overhang. The table saw sits against the far (north) edge.

10-final-look-sw.svg

FINAL LOOK - in use: miter saw flipped up, drop-leaves up, laser well open

The platform flipped: DWS779 deck flush with the top, hood behind it. Drop-leaves add 24" of support each side. The laser sits over the open well for rotary work.

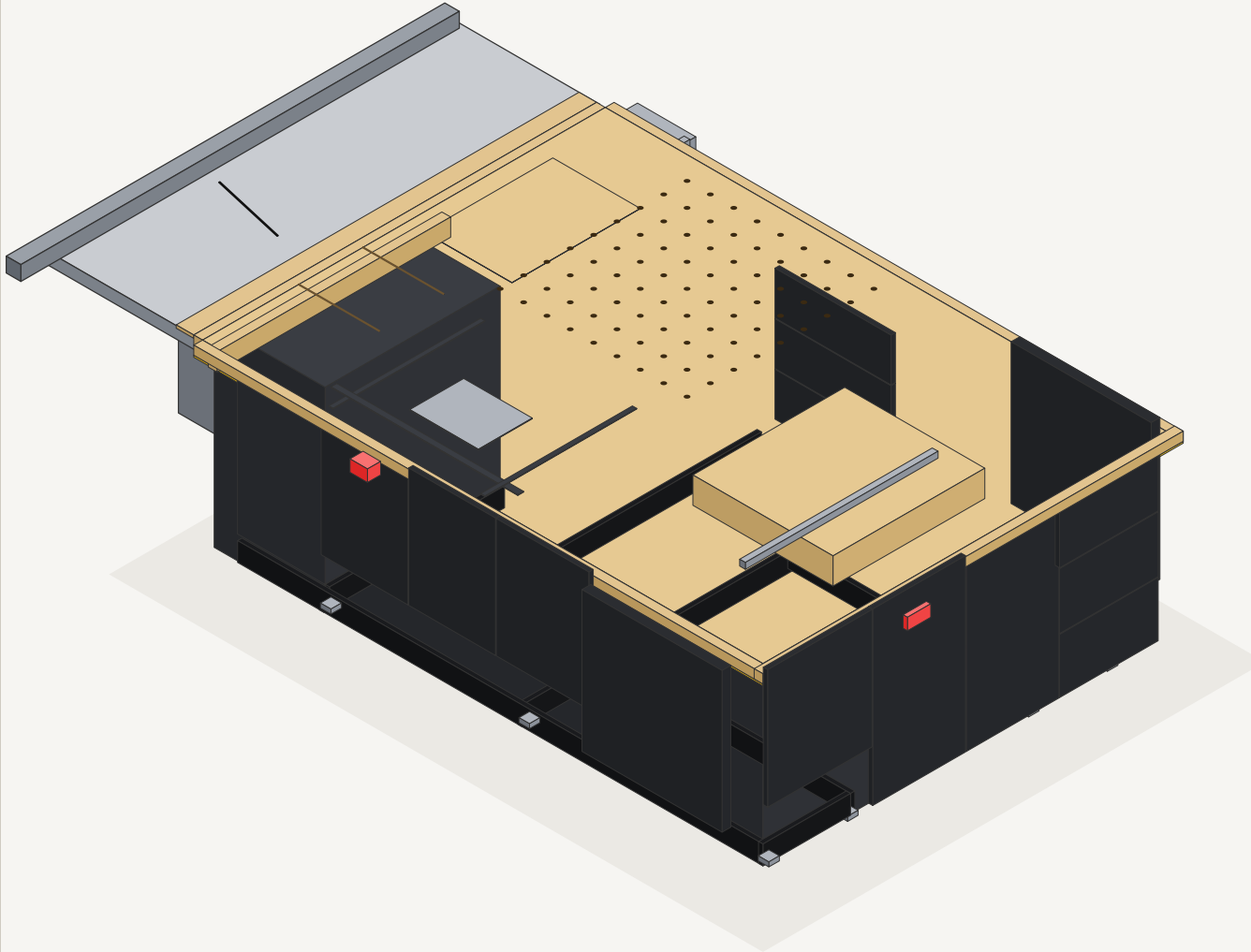


The flush box now hangs under the saw. Bay doors are closed again; the 4" quick-connect and the plug are made inside the bay.

11-final-look-in-use.svg

FINAL LOOK - from the north-east: router station face and outfeed edge

Mirrored view. East face: router-bit pull-out, gasketed router doors with paddle switch, utility column with the 4" and 2-1/2" ports and the 20 A inlet. North edge: bridge lip over the saw's rear rail.



The motor bay recess is behind the saw and not visible here; see Sheet 5 and Section C-C.

12-final-look-ne.svg

Design decisions

Design decisions — stage 2

This records what changed after your answers (transcribed in `answers-raw.txt`) and why. The stage-1 brief is kept as history in `00-design-brief.md`.

The one big change: no more L

You set a hard rule that the bench be no wider than the saw with its 36" extension (69-1/8"). That rule kills the stage-1 L-shape for a reason that isn't obvious until you draw the outfeed path: an 8' board being ripped travels the full length of the bench at table height. With only 69" of width, a rip between the blade and the fence occupies the middle of the top (x = 11" to 47" from the west edge), and any tool that stands above the top in that path gets hit. A miter saw in a wing beside the outfeed strip is exactly such a tool.

So the miter saw has to *disappear* when it is not in use. Your choice for that is a **flip-top platform**, the mechanism used on flip carts: a two-sided platform on a steel axle inside the base. The DWS779 bolts to one face; a hollow "flush box" the same height as the saw's deck (4-1/2") sits on the other. Open the two doors on the south face, release the locks, and rotate the platform half a turn: the saw swings out through the doors and comes up through a 29" x 24-1/2" hatch in the top with its deck flush at 34-3/4"; the box hangs underneath. Flip it back and the box's birch face closes the hatch and the whole 69" x 96" top is flat. No hydraulics, no leaves to store, a ten-second change.

The geometry that makes a tall slider like the DWS779 flip is in Sheet 7 and Sheet 10: the axle is 5-1/2" below the top so the stowed saw hangs to 6-3/4" off the floor, and the saw's top corners sweep a 25-1/2" radius that passes 3" in front of the bench's south edge, which is why the bay front is doors rather than a fixed panel. The safety system is four layers

deep (rest blocks that carry the load, index plungers, toggle clamps, and a seat-switch interlock on the saw's power) plus about 70 lb of steel in the box to balance the saw; see `07-stations.md`.

A hydraulic scissor lift is the alternative if you ever prefer a push-button rise over a flip; it needs the same bay and hatch length plus 42" of hatch, and the earlier sheets for it are in the repo history.

Concept: four faces, four stations

A 69" x 96" island, 34-3/4" tall, with the table saw docked against the north edge.

Face	Station	What is there
North	Outfeed	Flat 20" strip with two 14" grooves aligned to the miter slots; bridge lip over the saw's rear rail; a 30" x 15" recess in the base (the motor bay) for the CNS motor and its 4" dust hose.
West	Hand work / clamping station	9" quick-release face vise at the north end; 81-hole 3/4" dog field on 4" centres over a sealed downdraft plenum; W1 hand-tool drawers incl. a switched charging drawer; W2 holds the laser on a pull-out tray.
East	Router station	JessEm Rout-R-Lift II in a 9-1/4 x 11-3/4 plate recess, 12" from the edge; two fence T-tracks 26" apart; combo T-track/miter slot; sealed router box ducted to the spine; 2-1/2" vac port for the fence; vertical bit pull-out; two gasketed doors; paddle switch.
South	Miter station	DWS779 on the flip-top platform; hatch; two bay doors; drop-leaf wings 24" x 24" on both ends (46" of support each side with the bench); long-item drawers; clamp-rack pull-out.

Top, NW	Laser well	22" x 15" opening with a flush insert. Remove the insert and drop the floor panel onto cleats at 2", 4-1/2" or 7" below the top; rest the Falcon A1 over the opening for rotary work.
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Your answers, and what each one decided

Your answer	Decision
Shop 12' x 20', bench no wider than the saw, fixed, ceiling 7.5'	69" x 96" island; ten 1,000 lb levelling feet, no casters. Total floor with the saw and walking room: about 6' x 16'. Flipped-up saw tops out at 56-1/4", well under the ceiling. Keep 30" clear in front of the south face for the swing.
SawStop CNS, 36" T-Glide, 34-3/4" assembled height	Top at 34-3/4"; levelers give +/- 3/4". Blade lands 47-1/8" from the west edge. Measure the rear-rail height and the motor's projection before cutting the bay.
DeWalt DWS779	Flip core 28 x 24; hatch 29 x 24-1/2; axle at 29-1/4" off the floor; flush box 4-1/2" tall to match the deck. Verify the saw's locked-down height (design assumes 21-1/2"), deck height (4-1/2") and base depth (22"). The box height must equal the deck height exactly.
DW618 + JessEm Rout-R-Lift II (to be bought again)	Plate recess 9-1/4 x 11-3/4 x 3/8 with leveling screws; box 22 x 20 x 20-1/2 clears the lift. Your existing fence: check its clamp spacing against the 26" track spacing before cutting the dados.
HF collector (4") and shop vac + cyclone (2-1/2")	Two trunks side by side in the spine: 4" main and 2-1/2" vac. Five 4" gates, two 2-1/2" gates. See 05-dust-collection.md .
120 V, 20 A max, likes the power-strip approach	Single 20 A feed to a flush inlet, metal main strip with master switch, branch strips, flush USB strips on W/S/E faces, paddle switches at router and miter. See 06-electrical.md .
3/4" dogs, Rockler accessories	3/4" holes on a 4" grid; 2-1/4" of material at the holes (top + doubler) so dogs and holdfasts grip.

"vise"	One 9" quick-release face vise at the NW corner of the west face, jaw liners in maple. A tail/wagon vise was dropped to keep the top uninterrupted.
Clamping station + clamp rack	The dog field is the clamping station (dogs, holdfasts, Rockler clamps, dog-hole pipe-clamp cradles); the clamp rack is a 16" x 28" x 20" vertical pull-out at the south-east.
Everything flush at saw height	Yes, everywhere, including the miter deck when raised.
Top: cheap but ideal; MDF worked OK	3/4" Baltic birch over 3/4" MDF , glued, 1-1/2" maple edge, hardwax-oil or wipe-on poly finish; a third 3/4" BB layer (the doubler) under the dog field. About \$330 in sheet goods vs \$1,000+ for maple, flatter than solid wood, and the birch face drills and wears far better than MDF.
Style: different from the Shop Nation bench, more character and tech	The rising saw, the laser well, the downdraft dog field, charcoal fronts with natural birch edges under a maple-edged top, LED strip under the overhang, USB-C at every face.
Creality Falcon A1, 15 x 22 drop area	Well at the NW of the top, tray in W2 on 30" slides. Verify the laser's footprint (design allows 26 x 23 x 7-1/2). Its exhaust hose goes to a window, never into the dust system.

Key dimensions

Item	Value
Top	69" x 96" x 1-1/2", edges 1-1/2" maple; overhang 1" on E, S, W; flush at N
Height	34-3/4" finished; base 33-1/4" on a 4" plinth recessed 3"
Cabinet depth	40" on the west face (36" drawers), 22" on the east, 45" on the south
Hatch	29" x 24-1/2", 1/2" chamfer under the north and south edges; filled by the flush box when stowed
Motor bay	30" x 15" x full height, centred 49" from the west edge
Plenum	40" x 34" x 6", sealed, one 4" inlet on its east wall

Laser well	22" x 15", floor 2" / 4-1/2" / 7" below the top
Flip platform	2" core 28" x 24", 1" steel axle 29-1/4" off the floor at y = 22", flush box 26" x 24" x 4-1/2", four maple rest blocks at 28-1/4"
Spine duct	5" x 8" chase at floor level, 79" long, under the SE and E2 cabinets
Router plate	centre 12" from the east edge, 62" from the south edge
Drop-leaves	24" x 24", top-hinged, folding brackets

Things you must measure before cutting (the VERIFY list)

1. Table height of the CNS on its mobile base, and the height of the top of the rear rail. Decides the leveler setting and the bridge lip.
2. How far the motor housing projects behind the table at 0 and 45 degrees of tilt. The bay is 15" deep; deepen it if needed.
3. Distance from the blade to each miter slot (design uses 5-1/2"). Sets the outfeed grooves.
4. DWS779: overall height with the head locked down (sets the stowed clearance and the swing radius), deck height (sets the flush-box height exactly), base depth and the rails' rearmost point (sets where the rails end when stowed).
5. Your router fence's clamp spacing. Sets the fence-track spacing (26" in the drawings).
6. The Falcon A1's footprint and height. Sets the tray and the well.
7. If any of those differ, change `MITER_SAW` and `FLIP` in `tools/model.py` and re-run `tools/build_all.py`; every drawing, the swing-radius check and the cut list update.

Materials & hardware

Materials and hardware

Prices are 2026 US ballpark figures for planning. The sheet counts come from the nested cut diagrams (renders/cuts/), not from guesswork.

Sheet goods and lumber

Item	Qty	Use	Est. each	Est. total
3/4" Baltic birch, 4x8	12 (+1 spare)	carcasses, top upper layer, doubler, plenum, fronts, pull-outs, hood, flip core skins, flush box	\$95	\$1,235
1/2" Baltic birch, 4x8	2	drawer boxes, laser tray sides	\$70	\$140
1/4" Baltic birch, 4x8	2	cabinet backs, drawer bottoms	\$45	\$90
3/4" MDF, 4x8	2	top lower layer, drop-leaf cores	\$45	\$90
8/4 hard maple	~28 bf	edging, outfeed beam, bridge lip, cleats, jaw liners, clamp bars, cradles, rest blocks	\$9/bf	\$250
Subtotal				\$1,805

Cheaper option: cabinet-grade birch plywood (\$60/sheet) for the carcasses saves about \$400 but the edges won't look as good where they show (finger pulls, pull-out edges).

Hardware

Item	Qty	Notes	Est. total
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Levelling feet, M10, 1,000 lb, with T-nut plates	10	plinth	\$60
Full-extension ball-bearing slides, 36", 100 lb	6 pr	SW long drawers, W1 drawers	\$190
Full-extension slides, 30", 100 lb	1 pr	W2 drawer	\$30
Heavy-duty full-extension slides, 30", 250 lb (e.g. Accuride 9301 class)	3 pr	laser tray, clamp rack, bit pull-out	\$210
Concealed hinges, 110 deg, full overlay, soft-close	8	router doors, flip-bay doors	\$48
Piano hinge, 24" x 1-1/2", stainless	2	drop-leaves	\$30
Folding shelf brackets, 20", 300 lb pair	2 pr	drop-leaf supports	\$60
Hook-on panel clips / French cleat stock	-	band panels, utility cover	\$10
Foam weatherstrip tape 3/8"	1 roll	router doors, plenum clean-out	\$8
Silicone sealant	2 tubes	plenum, hood, router box	\$12
Threaded inserts 1/4-20 + bolts	8	saw to flip core, hood	\$15
1" cold-rolled steel rod, 36"	1	flip axle	\$25
1" 2-bolt flange bearings (UCFL205 class)	2	on the bay walls	\$30
1" shaft collars, set-screw	4	lock the core to the axle	\$20
1/2" spring index plungers (lock-out type)	2	locate the platform at 0 and 180 deg	\$30
Hold-down toggle clamps, 500 lb	2	clamp the core to the rest blocks	\$30

Plunger limit switch, 15 A rated (seat switch)	1	power interlock for the saw	\$15
Steel plate or flat bar, ~70 lb	-	counterweight inside the flush box (scrap yard)	\$60
4" quick-connect hose cuff + coupler	1	hood hose	\$20
Screws: #8 x 1-1/4, #8 x 2, #6 x 3/4; pocket screws 1-1/4	-		\$60
Wood glue (Titebond III), 1 gal	1	top lamination	\$35
Subtotal			\$983

Stations

Item	Qty	Notes	Est.
JessEm Rout-R-Lift II (3-1/2" motor, for the DW618)	1	plate 9-1/4 x 11-3/4 x 3/8	\$260
T-track, 3/4" x 3/8" aluminium, 48"	2	cut to 2 x 26" fence tracks	\$40
Combo T-track / miter-slot track, 36"	1	cut to 32"	\$45
Quick-release front vise, 9" (Eclipse / Rockler / WoodRiver)	1		\$150
Paddle safety switch, 120 V 15 A	2	router, miter	\$50
Bench dog / holdfast / Rockler accessories	-	you own these	-
Subtotal			\$545

Dust

Item	Qty	Est.
4" thin-wall PVC (S&D), 10'	2	\$40
4" 45-deg wye	5	\$60
4" 45-deg elbow	6	\$40

4" blast gate (metal, self-cleaning)	6	\$75
4" flex hose	15'	\$45
4" hose clamps, couplers, wall flange	-	\$40
2-1/2" PVC + 2 wyes + 4 elbows	-	\$35
2-1/2" blast gate	3	\$30
2-1/2" flex hose + 2-1/2" flip-lid port	10' + 1	\$35
Subtotal		\$400

Electrical

Item	Qty	Est.
Flush-mount 20 A inlet (L5-20 or straight-blade) + 12 AWG SJOOW cord, 15'	1	\$60
Metal surge power strip, 15 A, switched, 12-outlet	1	\$45
Metal branch power strips, 6-outlet	3	\$60
Flush-mount outlet strips with USB-A/C	3	\$105
Single-outlet switched box (charging drawer)	1	\$15
Surface raceway, 1" x 1/2", 20'	1	\$30
12 V LED strip, 5 m, warm white, + driver + PIR sensor	1	\$45
Subtotal		\$360

Finish

Item	Qty	Est.
Hardwax oil (Osmo / Rubio) or wipe-on poly, top and maple	1 qt	\$60
Charcoal enamel (satin) or black laminate for fronts and exposed base sides	1 gal	\$60
Dewaxed shellac, cabinet interiors	1 qt	\$25
Black paint, plinth	1 qt	\$15

Subtotal		\$160
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Grand total

Wood	\$1,805
Hardware (incl. flip-top axle, bearings, locks, counterweight)	\$983
Stations (router lift, vise, tracks, switches)	\$545
Dust	\$400
Electrical	\$360
Finish	\$160

Total	about \$4,250
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Without the router lift (if the JessEm purchase waits): about \$3,990. With cabinet-grade ply instead of Baltic birch: subtract about \$400.

Consumables and shop supplies

Sandpaper 80/120/180, blue tape, painter's plastic, clamps (you have those), a 3/4" brad-point or Forstner bit and a 3/4" dog-hole jig (or a drill guide), 1/2" and 1/4" router bits for the bit boards, a 3/4" straight bit and a 3/8" straight bit for the T-track dados, a 1/2" spiral for the plate recess, a 45-deg chamfer bit for the finger pulls.

Cut list

Cut list — every piece of wood

Generated by `tools/build_docs.py` from `tools/parts.py`; dimensions are finished sizes in inches (length x width; thickness is the material). Part IDs match the labels on the cut diagrams in `renders/cuts/`. Grain **L** means the length runs along the sheet's 96" axis; **A** means either way is fine.

Sheet and board count

Material	Sheets / boards	Cut diagram
3/4" Baltic birch 4x8	12	BB34-sheets-1 , BB34-sheets-2
1/2" Baltic birch 4x8	2	BB12-sheets-1
1/4" Baltic birch 4x8	3	BB14-sheets-1
3/4" MDF 4x8	2	MDF34-sheets-1
hard maple 8/4	about 28 board feet (incl. 30% waste); 10 pieces of 3" x 8' shown	MAPLE-boards

Buy one extra sheet of 3/4" Baltic birch: the nesting is tight (92-100% yield) and it leaves no room for a bad cut.

Parts by build step

Step 1 — Plinth and levelling feet

ID	Part	Qty	Length	Width	Material	Grain	Notes
PL - W0	plinth W rail	1	90"	4"	3/4" Baltic birch	L	4" tall rib, glued and screwed; toe recess 3"
PL - E1	plinth E rail	1	90"	4"	3/4" Baltic birch	L	4" tall rib, glued and screwed; toe recess 3"
PL - N2	plinth N rail (W of bay)	1	30"	4"	3/4" Baltic birch	L	4" tall rib, glued and screwed; toe recess 3"
PL - bay3	plinth bay S rail	1	30"	4"	3/4" Baltic birch	L	4" tall rib, glued and screwed; toe recess 3"
PL - S4	plinth S rail W	1	14"	4"	3/4" Baltic birch	L	4" tall rib, glued and screwed; toe recess 3"
PL - S5	plinth S rail E	1	15"	4"	3/4" Baltic birch	L	4" tall rib, glued and screwed; toe recess 3"
PL - flip6	plinth flip bay side rails	2	42"	4"	3/4" Baltic birch	L	4" tall rib, glued and screwed; toe recess 3"

PL - cross7	plinth cross rib y46	1	61"	4"	3/4" Baltic birch	L	4" tall rib, glued and screwed; toe recess 3"
PL - cross8	plinth cross rib y66	1	37"	4"	3/4" Baltic birch	L	4" tall rib, glued and screwed; toe recess 3"
PL - cross9	plinth cross rib y80	1	61"	4"	3/4" Baltic birch	L	4" tall rib, glued and screwed; toe recess 3"
PL - KICK	toe-kick face boards (W, S x2, E)	3	90"	4"	3/4" Baltic birch	L	paint black; one 90" board is cut into the two south pieces

Step 2 — Carcasses (base cabinets, spine, bays)

ID	Part	Qty	Length	Width	Material	Grain	Notes
SW - S	SW side	2	45"	29-1/4"	3/4" Baltic birch	L	cabinet SW: Long-item drawers
SW - TB	SW top/bottom	2	15-1/2"	45"	3/4" Baltic birch	A	
SW - BK	SW back	1	17"	29-1/4"	1/4" Baltic birch	A	overlay back, glued + screwed
SE - S	SE side	2	45"	20-1/2"	3/4" Baltic birch	L	cabinet SE: Clamp rack + bit pull-out
SE - TB	SE top/bottom	2	16-1/2"	45"	3/4" Baltic birch	A	
SE - BK	SE back	1	18"	20-1/2"	1/4" Baltic birch	A	overlay back, glued + screwed
SE - rack/bit divider	SE rack/bit divider	1	16-1/2"	20-1/2"	3/4" Baltic birch	A	
W1 - S	W1 side	2	40"	22-1/2"	3/4" Baltic birch	L	cabinet W1: Hand-tool drawers
W1 - TB	W1 top/bottom	2	18-1/2"	40"	3/4" Baltic birch	A	
W1 - BK	W1 back	1	20"	22-1/2"	1/4" Baltic birch	A	overlay back, glued + screwed
W2 - S	W2 side	2	33"	20"	3/4" Baltic birch	L	cabinet W2: Laser tray + drawer
W2 - TB	W2 top/bottom	2	27-1/2"	33"	3/4" Baltic birch	A	

W2 - BK	W2 back	1	29"	20"	1/4" Baltic birch	A	overlay back, glued + screwed
E2 - S	E2 side	2	22"	20-1/2"	3/4" Baltic birch	L	cabinet E2: Router cabinet
E2 - TB	E2 top/bottom	2	28-1/2"	22"	3/4" Baltic birch	A	
E2 - BK	E2 back	1	30"	20-1/2"	1/4" Baltic birch	A	overlay back, glued + screwed
E2 - router - box partition	E2 router-box partition	2	21-1/4"	20-1/2"	3/4" Baltic birch	A	
SP - WALL	spine duct inner wall (x=55)	1	79"	8"	3/4" Baltic birch	L	forms the 5 x 8 duct chase with the SE/E2 west sides
SP - LID	duct leg lid (y 76-80)	1	18"	4"	3/4" Baltic birch	A	
SP - LOW	SE/E2 lower side extension (z 4-12.75)	2	45"	8-3/4"	3/4" Baltic birch	L	closes the spine under the raised floors
BAY - S	motor bay south wall (y=80)	1	30"	29-1/4"	3/4" Baltic birch	A	
BAY - E	motor bay east wall (x=64)	1	15"	29-1/4"	3/4" Baltic birch	A	
BAY - NE	north-east filler panels (x 64-68)	2	15"	29-1/4"	3/4" Baltic birch	A	east and north faces of the 4" column
GAP - W	service-gap wall (x=41, y 66-80)	1	14"	22-1/2"	3/4" Baltic birch	A	supports the plenum floor
GAP - S	service-gap closure (y=46, x 41-46)	1	5"	29-1/4"	3/4" Baltic birch	A	
W - BAND1	west band panel over W1 (plenum face)	1	20"	6-3/4"	3/4" Baltic birch	A	with 6 x 4 plenum clean-out hatch
W - BAND2	west band panel over W2	1	29"	9-1/4"	3/4" Baltic birch	A	
E - UTIL	utility column cover (east face y 76-80)	1	4"	21-1/4"	3/4" Baltic birch	A	removable; ports and inlet mount here

E - BACK	east panel beside motor bay (y 80-95)	1	15"	29-1/4"	3/4" Baltic birch	A	
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Step 3 — Downdraft plenum, laser well and vise block

ID	Part	Qty	Length	Width	Material	Grain	Notes
PLN - F	plenum floor	1	40"	34"	3/4" Baltic birch	A	seal all joints with silicone
PLN - WL	plenum long walls	2	40"	5-1/4"	3/4" Baltic birch	A	
PLN - WS	plenum short walls	2	32-1/2"	5-1/4"	3/4" Baltic birch	A	east wall gets the 4" riser hole
PLN - SP	spacer frame over W2 (z 24-26.5)	4	33"	2-1/2"	3/4" Baltic birch	A	two long, two cut to 12"
WELL - WL	laser well walls (E-W)	2	23-1/2"	9-1/4"	3/4" Baltic birch	A	
WELL - WS	laser well walls (N-S)	2	15"	9-1/4"	3/4" Baltic birch	A	
WELL - FL	laser well floor panel (movable)	1	21-7/8"	14-7/8"	3/4" Baltic birch	A	rests on cleats at 2, 4-1/2 or 7 in. below the top
WELL - CL	well cleats	6	15"	3/4"	hard maple (8/4)	L	3 pairs, 3/4 x 3/4
WELL - LG	well insert ledge	2	37"	3/4"	hard maple (8/4)	L	3/4 x 3/4, 1-1/2 below the top surface
WISE - PAD	vise mounting pad (3 layers)	3	13"	12"	3/4" Baltic birch	A	laminated 2-1/4 thick under the top
WISE - JAW	vise jaw liners	2	9"	5"	hard maple (8/4)	L	

Step 7 — Laminate and machine the top

ID	Part	Qty	Length	Width	Material	Grain	Notes
TOP - U1	top upper layer, west piece (x 0-21)	1	96"	21"	3/4" Baltic birch	L	
TOP - U2	top upper layer, east piece (x 21-69)	1	96"	48"	3/4" Baltic birch	L	
TOP - L1	top lower layer, west piece (x 0-48)	1	96"	48"	3/4" MDF	L	
TOP - L2	top lower layer, east piece (x 48-69)	1	96"	21"	3/4" MDF	L	
TOP - DBL	doubler / plenum lid under dog field	1	40"	34"	3/4" Baltic birch	A	drill the dog holes through it with the top

Step 8 — Fit the top, outfeed lip and edging

ID	Part	Qty	Length	Width	Material	Grain	Notes
EDGE - W	maple edge band, west	1	99"	1-1/2"	hard maple (8/4)	L	mitred corners

EDGE - E	maple edge band, east	1	99"	1-1/2"	hard maple (8/4)	L	
EDGE - S	maple edge band, south	1	72"	1-1/2"	hard maple (8/4)	L	
EDGE - N	maple edge band, north (outfeed edge)	1	69"	1-1/2"	hard maple (8/4)	L	1/8 bevel on the top corner
LIP	bridge lip over the rear fence rail	1	69"	3"	hard maple (8/4)	L	MEASURE the rail first
BEAM	outfeed beam across the motor bay	1	40"	3"	hard maple (8/4)	L	glued + screwed under the top, x 28-68

Step 10 — Miter-saw flip-top station

ID	Part	Qty	Length	Width	Material	Grain	Notes
FLIP - SKIN	flip core skins (top and bottom)	2	24"	28"	3/4" Baltic birch	A	saw bolts through the upper one
FLIP - RIB	flip core ribs (1/2" between the skins)	4	27"	1/2"	1/2" Baltic birch	A	one is drilled for the 1" axle sleeve; glue and screw
FLIP - RIBL	flip core long ribs	2	24"	1/2"	1/2" Baltic birch	A	
BOX - TOP	flush box top (the visible face when stowed)	1	24"	26"	3/4" Baltic birch	A	cut from the hatch waste; birch face up
BOX - SIDE	flush box sides	2	24"	3-3/4"	3/4" Baltic birch	A	
BOX - END	flush box ends	2	24-1/2"	3-3/4"	3/4" Baltic birch	A	one end has a lift-off lid for the counterweight
HOOD - BK	hood back	1	28"	14"	3/4" Baltic birch	A	
HOOD - SD	hood sides	2	6"	14"	3/4" Baltic birch	A	
HOOD - FL	hood sloped floor + top	2	28"	6-1/2"	3/4" Baltic birch	A	floor slopes to the 4" quick-connect port
BEAR - BLK	bearing doubler blocks on the bay walls	4	12"	8"	3/4" Baltic birch	A	two per wall, laminated to 1-1/2, the flange bearings bolt through them
REST	rest blocks (platform seats on these, both ways up)	4	4"	2-1/2"	hard maple (8/4)	L	glued + screwed to the bay walls at 28-1/4" from the floor; one carries the seat switch

LEAF	drop-leaf wings (BB + MDF laminate)	2	24"	24"	3/4" Baltic birch	A	laminate to MDF LEAF-M; edge with maple
LEAF - M	drop-leaf wings, MDF layer	2	24"	24"	3/4" MDF	A	
LEAF - EDGE	drop-leaf edging	6	26"	1-1/2"	hard maple (8/4)	L	

Step 11 — Laser well and tray

ID	Part	Qty	Length	Width	Material	Grain	Notes
LT - BASE	laser tray base	1	30"	29"	3/4" Baltic birch	A	on 30" 100 lb full-extension slides
LT - SIDE	laser tray sides	2	30"	2"	1/2" Baltic birch	L	
LT - END	laser tray ends	2	28"	2"	1/2" Baltic birch	L	

Step 12 — Drawers, pull-outs and fronts

ID	Part	Qty	Length	Width	Material	Grain	Notes
D00R - BAY	flip-bay doors	2	15-7/8"	29-1/8"	3/4" Baltic birch	L	full overlay; open both before flipping; right one carries the paddle switch
LT - FR	laser tray front	1	28-7/8"	8-7/8"	3/4" Baltic birch	L	full overlay
SW - D3 - DS	SW-D3 long drawer (bottom) box side	2	36"	8"	1/2" Baltic birch	L	
SW - D3 - DF	SW-D3 long drawer (bottom) box front/back	2	15"	8"	1/2" Baltic birch	A	
SW - D3 - DB	SW-D3 long drawer (bottom) bottom	1	35-1/2"	15-1/2"	1/4" Baltic birch	A	in 1/4 groove
SW - D3 - FR	SW-D3 long drawer (bottom) front	1	16-7/8"	9-3/8"	3/4" Baltic birch	L	full overlay, 1/8 reveal; 30 deg finger-pull chamfer on top edge
SW - D2 - DS	SW-D2 long drawer box side	2	36"	8-1/4"	1/2" Baltic birch	L	
SW - D2 - DF	SW-D2 long drawer box front/back	2	15"	8-1/4"	1/2" Baltic birch	A	

SW-D2-DB	SW-D2 long drawer bottom	1	35-1/2"	15-1/2"	1/4" Baltic birch	A	in 1/4 groove
SW-D2-FR	SW-D2 long drawer front	1	16-7/8"	9-5/8"	3/4" Baltic birch	L	full overlay, 1/8 reveal; 30 deg finger-pull chamfer on top edge
SW-D1-DS	SW-D1 long drawer (top) box side	2	36"	8-1/2"	1/2" Baltic birch	L	
SW-D1-DF	SW-D1 long drawer (top) box front/back	2	15"	8-1/2"	1/2" Baltic birch	A	
SW-D1-DB	SW-D1 long drawer (top) bottom	1	35-1/2"	15-1/2"	1/4" Baltic birch	A	in 1/4 groove
SW-D1-FR	SW-D1 long drawer (top) front	1	16-7/8"	9-7/8"	3/4" Baltic birch	L	full overlay, 1/8 reveal; 30 deg finger-pull chamfer on top edge
CR-BASE	clamp rack base + top	2	28"	17"	3/4" Baltic birch	A	
CR-UP	clamp rack uprights	2	28"	17-1/2"	3/4" Baltic birch	A	notched for the bars
CR-BAR	clamp bars	4	15-1/2"	1-1/2"	hard maple (8/4)	L	F-clamps and bar clamps hang here
CR-FR	clamp rack front	1	17-7/8"	20-3/8"	3/4" Baltic birch	L	
BIT-BASE	bit pull-out base + top	2	12"	14"	3/4" Baltic birch	A	
BIT-UP	bit pull-out uprights	2	12"	17-1/2"	3/4" Baltic birch	A	
BIT-BRD	bit boards (drilled 1/4 and 1/2)	3	12"	17-1/2"	3/4" Baltic birch	A	holes at 1-1/2 spacing
BIT-FR	bit pull-out front	1	15-7/8"	20-3/8"	3/4" Baltic birch	L	
D00R-L	router cabinet door L (gasketed)	1	14-7/8"	20-3/8"	3/4" Baltic birch	L	full overlay; foam-tape gasket on the carcass
D00R-R	router cabinet door R (gasketed) + paddle switch	1	14-7/8"	20-3/8"	3/4" Baltic birch	L	full overlay; foam-tape gasket on the carcass

W1-D3-DS	W1-D3 box side	2	36"	6"	1/2" Baltic birch	L	
W1-D3-DF	W1-D3 box front/back	2	18"	6"	1/2" Baltic birch	A	
W1-D3-DB	W1-D3 bottom	1	35-1/2"	18-1/2"	1/4" Baltic birch	A	in 1/4 groove
W1-D3-FR	W1-D3 front	1	19-7/8"	7-3/8"	3/4" Baltic birch	L	full overlay, 1/8 reveal; 30 deg finger-pull chamfer on top edge
W1-D2-DS	W1-D2 box side	2	36"	6"	1/2" Baltic birch	L	
W1-D2-DF	W1-D2 box front/back	2	18"	6"	1/2" Baltic birch	A	
W1-D2-DB	W1-D2 bottom	1	35-1/2"	18-1/2"	1/4" Baltic birch	A	in 1/4 groove
W1-D2-FR	W1-D2 front	1	19-7/8"	7-3/8"	3/4" Baltic birch	L	full overlay, 1/8 reveal; 30 deg finger-pull chamfer on top edge
W1-D1-DS	W1-D1 charging drawer box side	2	36"	6"	1/2" Baltic birch	L	
W1-D1-DF	W1-D1 charging drawer box front/back	2	18"	6"	1/2" Baltic birch	A	
W1-D1-DB	W1-D1 charging drawer bottom	1	35-1/2"	18-1/2"	1/4" Baltic birch	A	in 1/4 groove
W1-D1-FR	W1-D1 charging drawer front	1	19-7/8"	7-3/8"	3/4" Baltic birch	L	full overlay, 1/8 reveal; 30 deg finger-pull chamfer on top edge
W2-D2-DS	W2-D2 box side	2	30"	9-1/2"	1/2" Baltic birch	L	
W2-D2-DF	W2-D2 box front/back	2	27"	9-1/2"	1/2" Baltic birch	A	
W2-D2-DB	W2-D2 bottom	1	29-1/2"	27-1/2"	1/4" Baltic birch	A	in 1/4 groove
W2-D2-FR	W2-D2 front	1	28-7/8"	10-7/8"	3/4" Baltic birch	L	full overlay, 1/8 reveal; 30 deg finger-pull chamfer on top edge

Step 13 — Vise, dogs, lighting and finish

ID	Part	Qty	Length	Width	Material	Grain	Notes
CRADLE	pipe-clamp cradles (dog-hole mounted)	6	6"	3"	hard maple (8/4)	L	3/4 dowel peg glued in; notch for 3/4 pipe

Parts by material (for the saw)

3/4" Baltic birch

ID	Qty	Length	Width	Part
TOP-U2	1	96"	48"	top upper layer, east piece (x 21-69)
TOP-U1	1	96"	21"	top upper layer, west piece (x 0-21)
PL-W0	1	90"	4"	plinth W rail
PL-E1	1	90"	4"	plinth E rail
PL-KICK	3	90"	4"	toe-kick face boards (W, S x2, E)
SP-WALL	1	79"	8"	spine duct inner wall (x=55)
PL-cross7	1	61"	4"	plinth cross rib y46
PL-cross9	1	61"	4"	plinth cross rib y80
SW-S	2	45"	29-1/4"	SW side
SE-S	2	45"	20-1/2"	SE side
SP-LOW	2	45"	8-3/4"	SE/E2 lower side extension (z 4-12.75)
PL-flip6	2	42"	4"	plinth flip bay side rails
PLN-F	1	40"	34"	plenum floor
TOP-DBL	1	40"	34"	doubler / plenum lid under dog field
W1-S	2	40"	22-1/2"	W1 side
PLN-WL	2	40"	5-1/4"	plenum long walls
PL-cross8	1	37"	4"	plinth cross rib y66
W2-S	2	33"	20"	W2 side
PLN-SP	4	33"	2-1/2"	spacer frame over W2 (z 24-26.5)

PLN-WS	2	32-1/2"	5-1/4"	plenum short walls
BAY-S	1	30"	29-1/4"	motor bay south wall (y=80)
LT-BASE	1	30"	29"	laser tray base
PL-N2	1	30"	4"	plinth N rail (W of bay)
PL-bay3	1	30"	4"	plinth bay S rail
W-BAND2	1	29"	9-1/4"	west band panel over W2
W2-D2-FR	1	28-7/8"	10-7/8"	W2-D2 front
LT-FR	1	28-7/8"	8-7/8"	laser tray front
E2-TB	2	28-1/2"	22"	E2 top/bottom
CR-UP	2	28"	17-1/2"	clamp rack uprights
CR-BASE	2	28"	17"	clamp rack base + top
HOOD-BK	1	28"	14"	hood back
HOOD-FL	2	28"	6-1/2"	hood sloped floor + top
W2-TB	2	27-1/2"	33"	W2 top/bottom
BOX-END	2	24-1/2"	3-3/4"	flush box ends
FLIP-SKIN	2	24"	28"	flip core skins (top and bottom)
BOX-TOP	1	24"	26"	flush box top (the visible face when stowed)
LEAF	2	24"	24"	drop-leaf wings (BB + MDF laminate)
BOX-SIDE	2	24"	3-3/4"	flush box sides
WELL-WL	2	23-1/2"	9-1/4"	laser well walls (E-W)
E2-S	2	22"	20-1/2"	E2 side
WELL-FL	1	21-7/8"	14-7/8"	laser well floor panel (movable)
E2-router-box partition	2	21-1/4"	20-1/2"	E2 router-box partition
W-BAND1	1	20"	6-3/4"	west band panel over W1 (plenum face)
W1-D3-FR	1	19-7/8"	7-3/8"	W1-D3 front
W1-D2-FR	1	19-7/8"	7-3/8"	W1-D2 front

W1-D1-FR	1	19-7/8"	7-3/8"	W1-D1 charging drawer front
W1-TB	2	18-1/2"	40"	W1 top/bottom
SP-LID	1	18"	4"	duct leg lid (y 76-80)
CR-FR	1	17-7/8"	20-3/8"	clamp rack front
SW-D1-FR	1	16-7/8"	9-7/8"	SW-D1 long drawer (top) front
SW-D2-FR	1	16-7/8"	9-5/8"	SW-D2 long drawer front
SW-D3-FR	1	16-7/8"	9-3/8"	SW-D3 long drawer (bottom) front
SE-TB	2	16-1/2"	45"	SE top/bottom
SE-rack/bit divider	1	16-1/2"	20-1/2"	SE rack/bit divider
D00R-BAY	2	15-7/8"	29-1/8"	flip-bay doors
BIT-FR	1	15-7/8"	20-3/8"	bit pull-out front
SW-TB	2	15-1/2"	45"	SW top/bottom
BAY-E	1	15"	29-1/4"	motor bay east wall (x=64)
BAY-NE	2	15"	29-1/4"	north-east filler panels (x 64-68)
E-BACK	1	15"	29-1/4"	east panel beside motor bay (y 80-95)
WELL-WS	2	15"	9-1/4"	laser well walls (N-S)
PL-S5	1	15"	4"	plinth S rail E
D00R-L	1	14-7/8"	20-3/8"	router cabinet door L (gasketed)
D00R-R	1	14-7/8"	20-3/8"	router cabinet door R (gasketed) + paddle switch
GAP-W	1	14"	22-1/2"	service-gap wall (x=41, y 66-80)
PL-S4	1	14"	4"	plinth S rail W
WISE-PAD	3	13"	12"	vise mounting pad (3 layers)
BIT-UP	2	12"	17-1/2"	bit pull-out uprights
BIT-BRD	3	12"	17-1/2"	bit boards (drilled 1/4 and 1/2)
BIT-BASE	2	12"	14"	bit pull-out base + top
BEAR-BLK	4	12"	8"	bearing doubler blocks on the bay walls

H00D - SD	2	6"	14"	hood sides
GAP - S	1	5"	29-1/4"	service-gap closure (y=46, x 41-46)
E - UTIL	1	4"	21-1/4"	utility column cover (east face y 76-80)

1/4" Baltic birch

ID	Qty	Length	Width	Part
W1 - D3 - DB	1	35-1/2"	18-1/2"	W1-D3 bottom
W1 - D2 - DB	1	35-1/2"	18-1/2"	W1-D2 bottom
W1 - D1 - DB	1	35-1/2"	18-1/2"	W1-D1 charging drawer bottom
SW - D3 - DB	1	35-1/2"	15-1/2"	SW-D3 long drawer (bottom) bottom
SW - D2 - DB	1	35-1/2"	15-1/2"	SW-D2 long drawer bottom
SW - D1 - DB	1	35-1/2"	15-1/2"	SW-D1 long drawer (top) bottom
E2 - BK	1	30"	20-1/2"	E2 back
W2 - D2 - DB	1	29-1/2"	27-1/2"	W2-D2 bottom
W2 - BK	1	29"	20"	W2 back
W1 - BK	1	20"	22-1/2"	W1 back
SE - BK	1	18"	20-1/2"	SE back
SW - BK	1	17"	29-1/4"	SW back

hard maple (8/4)

ID	Qty	Length	Width	Part
EDGE - W	1	99"	1-1/2"	maple edge band, west
EDGE - E	1	99"	1-1/2"	maple edge band, east
EDGE - S	1	72"	1-1/2"	maple edge band, south
LIP	1	69"	3"	bridge lip over the rear fence rail
EDGE - N	1	69"	1-1/2"	maple edge band, north (outfeed edge)
BEAM	1	40"	3"	outfeed beam across the motor bay

WELL - LG	2	37"	3/4"	well insert ledge
LEAF - EDGE	6	26"	1-1/2"	drop-leaf edging
CR - BAR	4	15-1/2"	1-1/2"	clamp bars
WELL - CL	6	15"	3/4"	well cleats
WISE - JAW	2	9"	5"	vise jaw liners
CRADLE	6	6"	3"	pipe-clamp cradles (dog-hole mounted)
REST	4	4"	2-1/2"	rest blocks (platform seats on these, both ways up)

3/4" MDF

ID	Qty	Length	Width	Part
T0P - L1	1	96"	48"	top lower layer, west piece (x 0-48)
T0P - L2	1	96"	21"	top lower layer, east piece (x 48-69)
LEAF - M	2	24"	24"	drop-leaf wings, MDF layer

1/2" Baltic birch

ID	Qty	Length	Width	Part
SW - D1 - DS	2	36"	8-1/2"	SW-D1 long drawer (top) box side
SW - D2 - DS	2	36"	8-1/4"	SW-D2 long drawer box side
SW - D3 - DS	2	36"	8"	SW-D3 long drawer (bottom) box side
W1 - D3 - DS	2	36"	6"	W1-D3 box side
W1 - D2 - DS	2	36"	6"	W1-D2 box side
W1 - D1 - DS	2	36"	6"	W1-D1 charging drawer box side
W2 - D2 - DS	2	30"	9-1/2"	W2-D2 box side
LT - SIDE	2	30"	2"	laser tray sides
LT - END	2	28"	2"	laser tray ends
W2 - D2 - DF	2	27"	9-1/2"	W2-D2 box front/back
FLIP - RIB	4	27"	1/2"	flip core ribs (1/2" between the skins)

FLIP-RIBL	2	24"	1/2"	flip core long ribs
W1-D3-DF	2	18"	6"	W1-D3 box front/back
W1-D2-DF	2	18"	6"	W1-D2 box front/back
W1-D1-DF	2	18"	6"	W1-D1 charging drawer box front/back
SW-D1-DF	2	15"	8-1/2"	SW-D1 long drawer (top) box front/back
SW-D2-DF	2	15"	8-1/4"	SW-D2 long drawer box front/back
SW-D3-DF	2	15"	8"	SW-D3 long drawer (bottom) box front/back

Build guide

Build guide — step by step

Thirteen steps in build order. Each step lists the parts (IDs from `03-cut-list.md`), the tools you need for that step, the procedure, a check before moving on, and the drawing to have open. The isometric step views are `renders/601-step-01.svg` through `613-step-13.svg`; the dimensioned sheets are `renders/20-` to `51-`.

Conventions: all cabinet joints are glue + #8 x 1-1/4" screws in countersunk holes (pocket screws where a face is hidden). Sand everything to 180 before assembly. Finish interiors with two coats of shellac before the fronts go on.

General shop tools used throughout: table saw with a good 80T blade for plywood, track saw or circular saw with a straightedge for breaking down sheets, router (the DW618 works), random-orbit sander, drill/driver, countersink, squares, tape, pencil, a 4' level, clamps, dust mask.

Step 1 — Plinth and levelling feet

Drawing: `601-step-01.svg`, *Sheet 1 (plan)*, *Sheet 6*.

Parts: PL-* ribs, PL-KICK toe-kick boards, 10 levelling feet.

Tools: table saw, miter saw (rip 4" strips, cross-cut), drill, 3/8" bit for the T-nut plates, level.

1. Rip 4"-wide strips of 3/4" Baltic birch and cross-cut the ribs to the lengths in the cut list.
2. Lay out the frame on the floor where the bench will live: outer rails at 4" and 65" from the west edge, cross ribs at y = 46, 66 and 80; the front rail is split (x 4-18 and 50-65) to leave the flip bay open; two 42" rails run north-south at x = 18 and x = 50 to carry the bay walls.

3. Glue and screw the ribs into a ladder. Drive T-nut plates for the ten levelers into the rib bottoms at the positions in the step view (corners, mid-rails, and both sides of the bay).
4. Screw the toe-kick faces on (west, east, and the two south pieces). Paint black.
5. Set the plinth in place and level it in both directions with a 4' level. This is the reference for everything above; take your time.

Check: diagonals equal within 1/16"; top of the plinth is level and 4" off the floor at every leveler.

Step 2 — Carcasses

Drawing: `602-step-02.svg`, *Sheet 6, 7, 8 (sections)*.

Parts: SW-, SE-, W1-, W2-, E2-* sides/tops/bottoms/back and dividers; SP-WALL, SP-LID, SP-LOW; BAY-S, BAY-E, BAY-NE; GAP-W, GAP-S; W-BAND1/2; E-UTIL; E-BACK.

Tools: track saw or table saw, router with a 1/4" straight bit (optional 1/4" dado for backs), drill, countersink, pocket-hole jig, clamps, square.

1. Cut all carcass parts. Label each with its ID in pencil on an edge that will be hidden.
 2. Build the five boxes one at a time on a flat surface: sides full height, top and bottom between the sides, back glued and screwed to the rear edges. Square each box against the back before the glue sets.
- **SW** (17 x 45 x 29-1/4, opens south). Its east side is the west wall of the flip bay: glue the two BEAR-BLK doubler blocks (Step 10) to it now, centred 22" from the south end and 29-1/4" from the floor, so the bearing bolts have 2-1/4" of material.
 - **SE** (18 x 45 x 20-1/2, opens south and east): its floor sits at z = 12-3/4 (raised over the spine); add the SP-LOW extension below the

sides and the rack/bit divider at 29" from the south end. Its west side is the east wall of the flip bay: doubler blocks as for SW.

- **W1** (20 wide x 40 deep x 22-1/2 tall, opens west).
 - **W2** (29 wide x 33 deep x 20 tall, opens west).
 - **E2** (30 wide x 22 deep x 20-1/2 tall, opens east): floor at $z = 12\text{-}3/4$; add the two router-box partitions at 6" and 25-1/4" from the south side, making a 20"-wide sealed compartment. Cut the 4" dust hole in the floor of that compartment at $x = 52\text{-}1/2$ (from the west edge of the bench) and a 1" x 4" make-up-air slot low on the north partition.
1. Cut the spine wall, motor-bay walls, service-gap walls and band panels. Leave them loose; they go on in Step 4.

Check: every box square, backs flush, faces flat. Dry-fit the five boxes on the plinth in their positions (Sheet 1) and confirm the gaps: 5" service gap between W1 and E2, 32" flip bay between SW and SE, 30" x 15" motor bay at the north-centre.

Step 3 — Downdraft plenum, laser well and vise block

Drawing: 603-step-03.svg, Sheet 6, Sheet 7.

Parts: PLN-F, PLN-WL x2, PLN-WS x2, PLN-SP x4; WELL-WL x2, WELL-WS x2, WELL-FL, WELL-CL x6, WELL-LG x2; VISE-PAD x3, VISE-JAW x2.

Tools: table saw, hole saw 4-1/8" (or jigsaw), caulk gun, clamps.

1. Plenum: build a 40 x 34 box, 6" tall, walls on top of the floor. Cut the 4" inlet hole in the east wall, centred 70" from the south edge of the bench and 2-1/2" above the plenum floor. Run silicone along every inside joint. The lid is the doubler (Step 7), so leave the top open for now.
2. Spacer frame: four 2-1/2" ribs that lift the plenum's north end over W2 (whose top is at 24").

3. Laser well: a 22 x 15 box (inside) 9-1/4" tall with no top. Glue the three pairs of cleats inside at 2", 4-1/2" and 7" below where the top surface will be (i.e. at 32-3/4, 30-1/4 and 27-3/4 from the floor). Glue the 3/4" insert ledge 1-1/2" below the top surface line. Cut the floor panel to drop onto the cleats with 1/16" clearance.
4. Vise pad: laminate three 12 x 13 pieces of 3/4" BB into a 2-1/4" block. Plane the jaw liners to fit your vise.

Check: plenum holds air (tape the inlet and open top, push on a wall: no hiss at the joints). Well floor drops onto all three cleat heights without binding.

Step 4 — Assemble the base

Drawing: 604-step-04.svg, Sheet 1, Sheets 2-5.

Tools: drill, clamps, level, 2" screws, helper.

1. Set SW and SE on the plinth at the south end, exactly 32" apart and parallel (the flip axle runs between them, so measure at both ends), backs to the north. Screw them to the plinth ribs from inside.
2. Set W1 north of SW (its south side is the flip-bay north wall for $x = 1\text{-}18$), and W2 north of W1. Set E2 north of SE. Screw adjacent boxes to each other through their sides with 1-1/4" screws.
3. Spine: stand SP-WALL at $x = 55$ between the SE/E2 west sides and the plinth; screw the SP-LID over the leg ($y = 76\text{-}80$). The 5 x 8 chase should now run from the south toe kick to the utility corner.
4. Motor bay: fit BAY-S ($y = 80$, $x = 34\text{-}64$) and BAY-E ($x = 64$); add the BAY-NE fillers so the north-east column is closed.
5. Service gap: GAP-W ($x = 41$, $y = 66\text{-}80$) and GAP-S ($y = 46$, $x = 41\text{-}46$).
6. Set the plenum on W1/W2 with the spacer frame under its north end; the inlet hole faces the service gap. Silicone the plenum floor to the cabinet tops. Set the laser well box on W2 with its west wall at $x = 12$ and south wall at $y = 78$. Glue the vise pad under where the top will

be, in the NW corner (x 1-13, y 82-95); it hangs on the W2 top and the well's west wall.

7. Hang the west band panels and the utility cover on hook-on clips so they can come off for service. The flip bay stays open at the south until its doors go on in Step 12.

Check: the whole base is level and its top edges are all at 33-1/4" from the floor (26-1/2" at W1, 24" at W2, plenum top at 32-1/2"). Measure diagonally corner to corner across the base: equal within 1/8".

Step 5 — Dust ducting

Drawing: 605-step-05.svg, Sheet 8, Sheet 11, 05-dust-collection.md.

Parts: 4" PVC, five 4" wyes, elbows, six 4" gates, 2-1/2" PVC and gates, flex hose, wall flange.

Tools: PVC saw or miter saw, deburring tool, hole saw 4-1/8" and 2-5/8", screwdriver, silicone.

1. Dry-fit the 4" main in the spine from the south toe kick to the leg, then east to the utility column. Place a wye for each drop at the y positions on Sheet 11 (floor sweep at the south end, miter hood at 30", router at 62", plenum at 70", table saw at 78"). The wye's branch must point toward the ports.
2. Add the 2-1/2" vac trunk beside it (above, in the 8" chase) with its two wyes (miter chute at 26", router fence at 64").
3. Glue the runs. Fit a blast gate at every branch, mounted where you can reach it: floor sweep behind a flap in the toe kick; miter hood and chute gates on the flip bay's east wall, low, with the 4" quick-connect cuff and the 2-1/2" cuff right at the gates; router gate on the box floor; plenum gate in the service gap (reach from the removed W-BAND1 clean-out); saw gate on the bay's south wall.
4. Run the plenum riser (4" PVC, two 45s) up the service gap into the plenum's east wall inlet. Silicone it.

5. Leave 4' of 4" flex and 3' of 2-1/2" flex coiled in the flip bay for the hood and the saw chute; they connect to the cuffs after each flip (Step 10). Leave 4' of 4" flex at the motor-bay gate for the CNS port.
6. Fit the wall flanges at the utility column: 4" at 6" above the floor, 2-1/2" above it.

Check: with the collector connected and one gate open at a time, feel strong suction at every drop. No leaks at the plenum (hold a tissue at the seams).

Step 6 — Electrical

Drawing: 606-step-06.svg, Sheet 12, 06-electrical.md.

Tools: drill, hole saw for the inlet, screwdriver, cable ties, multimeter.

1. Mount the flush inlet in the utility cover (E-UTIL) at 18" above the floor and the main strip on the inside of the column.
2. Run the raceway along the spine lid (top of the chase, not inside it) from the column to the south end, and a branch through the service gap to the west.
3. Mount tool outlets: a single outlet inside the router box (high, away from the bit) with a paddle switch on the right-hand door front; a single outlet on the flip bay's east wall at 12" height, wired **through the seat switch** (Step 10) and a paddle switch on the right-hand bay door; the switched outlet for the charging drawer inside W1; an outlet inside W2 for the laser.
4. Mount the flush USB strips: west face under the top overhang at y 50-58; south face at x 56-64; east face at y 68-74. Mount the saw outlet on the north face beside the bay.
5. LED strip in the top overhang (after Step 8) with the driver on the main strip and a PIR sensor at the south face.
6. Plug everything into the main strip; label the cords.

Check: with the main switch on, every outlet reads 120 V; the paddle switches kill their tool; the master kills all.

Step 7 — Laminate and machine the top

Drawing: [607-step-07.svg](#), Sheet 1, Sheet 9, Sheet 10.

Parts: TOP-U1, TOP-U2 (3/4" BB), TOP-L1, TOP-L2 (3/4" MDF), TOP-DBL.

Tools: glue roller, every clamp you own plus cauls, track saw, router with 1/2" spiral, 3/4" and 3/8" straight bits, dog-hole jig with a 3/4" bit, jigsaw, pattern bit, plunge base, straightedge.

1. Lay the two MDF pieces on a dead-flat surface (the base is flat: cover it with plastic and use it) with the seam at x = 48. Lay the birch pieces on top with the seam at x = 21, so the seams do not line up. Glue the layers with a roller, screw temporarily from below through the MDF every 8", and clamp the edges. Let it cure overnight; remove the screws.
2. Trim the top to 69 x 96 with the track saw.
3. Glue the doubler under the dog-field area (x 1-41, y 46-80).
4. Lay out from the north-west corner. Cut the flip hatch (29 x 24-1/2 at x 19-1/2, y 9-3/4) with the track saw plunge-cutting and finish the corners with a jigsaw; keep the cut-out, its birch face becomes the top of the flush box. Rout a 1/2" 45-degree chamfer on the *underside* of the hatch's north and south edges: the box corners sweep past there. Cut the laser well opening (22 x 15 at x 12, y 78); keep the cut-out as the insert. Cut the router plate opening: rout the 3/8"-deep rabbet to the plate's outline with a pattern bit against a template, then cut through the inner opening 1" smaller all round.
5. Rout the dados: two 3/4" x 3/8" for the fence T-tracks at y 49 and y 75 from x 42 to 68; the combo track dado at x 64-1/2 from y 46 to 78 (check the track's width); two outfeed grooves 3/4" x 3/8" x 14" at x 41-5/8 and 52-5/8 from the north edge (verify against your miter slots).
6. Drill the 81 dog holes with the jig: 3/4" on 4" centres, x 4-36, y 46-78. Chamfer both ends. Drill through the top and doubler; they open into

the plenum.

7. Fit the well ledge and test the insert.

Check: top flat within 1/32" over its length (winding sticks); insert flush.

Step 8 — Fit the top, outfeed lip and edging

Drawing: [608-step-08.svg](#), Sheet 5, Sheet 8.

Parts: EDGE-W, EDGE-E, EDGE-S, EDGE-N, LIP, BEAM.

Tools: helper (the top is about 220 lb), miter saw, biscuit joiner or dominos (optional), clamps, block plane, router with roundover and chamfer bits.

1. Glue and screw the outfeed beam under the top's north edge from x 28 to 68 (it bridges the motor bay).
2. Lift the top onto the base, north edge flush with the base's north face, 1" overhang on the other three sides. Screw up through the cabinet tops and the plenum lid with 1-1/4" screws in oversized holes (the top must be able to move a hair). Bolt the vise pad area with 2" screws.
3. Mitre the maple edging, glue it on with biscuits or splines. Flush it with a block plane. Ease the E/S/W edges with a 3/16" roundover; put a 1/8" bevel on the north edge's top corner only.
4. Bridge lip: measure the gap between the saw's rear-rail face and the cast table, and the rail's height. Cut the lip to bridge it with 1/16" of clearance over the rail and screw it to the north edge band at the top surface level.
5. Roll the saw up to the bench. Adjust the levelers until the top sits 1/16" below the saw table everywhere along the north edge and the grooves align with the miter slots.

Check: a straightedge from the saw table onto the outfeed drops 1/16" and never catches. A sled runs off the saw into the grooves.

Step 9 — Router station

Drawing: 609-step-09.svg, Sheet 9.

Tools: drill, screwdriver, straightedge, feeler gauge.

1. Install the T-tracks in their dados with #6 x 3/4" screws every 6"; the combo track too.
2. Set the Rout-R-Lift II plate into its rabbet, level it with the corner screws until it is dead flush with the top all round.
3. Mount the DW618 motor in the lift; plug it into the outlet in the box; route the cord through a grommet in the partition.
4. Drill a 2-5/8" hole in the east edge band at y 62, 31" high, for the flip-lid vac port; connect the 2-1/2" branch behind it.
5. Foam-tape the door openings; hang the two doors on their hinges; fit the paddle switch to the right-hand door front, wired to the box outlet.
6. Clamp your fence into the two tracks and check it is square to the combo track.

Check: plate flush; bit raises and lowers freely through the plate; with the collector on and the box gate open, chips at a test cut go down, not out.

Step 10 — Miter-saw flip-top station

Drawing: 610-step-10.svg, Sheet 7 (three states), Sheet 10 (plan and safety).

Parts: FLIP-SKIN x2, FLIP-RIB x4, FLIP-RIBL x2 (core); BOX-TOP, BOX-SIDE x2, BOX-END x2 (flush box); HOOD-*; BEAR-BLK x4; REST x4; LEAF x2 + LEAF-M x2 + LEAF-EDGE; 1" rod, 2 flange bearings, 4 shaft collars, 2 index plungers, 2 toggle clamps, seat switch, ~70 lb steel plate, 1/4-20 inserts, piano hinges, folding brackets.

Tools: drill press or a careful drill with a 1" bit (the axle holes must be true), wrenches, level, 4' straightedge, clamps, a helper for the first flips.

1. **Core.** Glue the 1/2" ribs between the two 3/4" skins to make a 28 x 24 x 2" torsion core. The middle rib runs E-W on the axle line and is drilled 1" for the rod before assembly; the end ribs close the box. Bore the two skins at the axle line too, so the rod passes straight through the core with 1/8" of skin bearing on each side of every rib.
2. **Axle.** Drill the bearing bolt holes in the doubler blocks on both bay walls with the walls clamped together as a pair, or use a jig, so the two bearings are coaxial: 29-1/4" above the floor, 22" from the south face. Bolt the flange bearings on. Slide the rod through one bearing, the core and the other bearing; centre the core; lock it with a shaft collar on each side of the core and one outside each bearing.
3. **Rest blocks and locks.** Glue and screw the four maple rest blocks to the bay walls with their tops at 28-1/4" (the core's underside when horizontal), 11" north and south of the axle. Spin the core to horizontal both ways and check it seats on all four. Mount the two index plungers on the walls 6" south of the axle so their pins enter 1/2" holes drilled in the core's edges at both the 0 and 180 degree positions. Mount the two toggle clamps on the walls 6" north of the axle so their pads press the core down onto the rest blocks with the platform seated either way (the core is symmetric about the axle, so one setting serves both).
4. **Seat switch.** Let a plunger limit switch into the top of one rest block so the core's underside depresses it only when fully seated. Wire the flip-bay outlet through it in series with the paddle. Test: the saw has no power until the platform is on the blocks and clamped.
5. **Flush box.** Build the 26 x 24 x 4-1/2 box (sides, ends, the birch-faced top from the hatch cut-out) and screw it to the core's lower face, centred. One end gets a lift-off lid. It stands 4-1/2" off the core: **this must equal the saw's deck height**; shim or trim to match your measurement.
6. **Saw and hood.** Flip the platform so the bare face is up (box down). Set the DWS779 on it with its fence line 11" north of the platform's south edge and centred E-W; mark and drill for 1/4-20 inserts; bolt it

down. Bolt the hood to the north 6" of the core behind the rails; fit the 4" quick-connect on its port.

7. **Balance.** Load the flush box with steel plate (start at 60 lb) until the platform, unlocked and held at 90 degrees, needs about one hand to move either way. Screw the plate down so it cannot shift. Fit the lid.
8. **Set the height.** With the saw up and clamped, lay a straightedge from the bench top across the saw's deck in both directions. Shim between the saw and the core, or plane the rest blocks, until it is flush within 1/32". Flip it, and check the box face the same way.
9. **Drop-leaves.** Laminate each 24 x 24 leaf (BB over MDF), edge with maple, hang on the piano hinges under the top's overhang at y 4-28 on both ends so the leaf's top face is flush when raised; fit the folding brackets to the base sides beneath.
10. **Swing check.** Lock the saw's head down, slide and bevel. Open the bay (no doors yet), stand clear, release the plungers and clamps and flip slowly ten times. Nothing may touch the top's edges, the hatch, the walls or the floor at any angle. Then fit the doors (Step 12) and confirm they clear the stowed saw.

Check: the platform seats on all four blocks both ways with the plungers in and the clamps closed; the saw deck and the box face are each flush; the seat switch kills the saw when a clamp is opened; the hose and cord reach their cuffs with the saw up.

Step 11 — Laser well and tray

Drawing: [611-step-11.svg](#), Sheet 7.

Parts: LT-BASE, LT-SIDE x2, LT-END x2, LT-FR, well insert (from Step 7), WELL-FL.

Tools: drill, screwdriver.

1. Build the tray (30 x 29 base, 2" sides) and hang it in W2 on the 30" heavy-duty slides with its top face at 15-3/4" from the floor. The laser stores on it; a strip of anti-slip mat keeps it put.

2. Fit the well insert (flush) and the floor panel (on the cleats). Test the rotary setup: insert out, floor at 7", laser resting over the opening on the top.
3. Cord: the laser plugs into the W2 outlet; the exhaust hose runs out to a window, never into the bench.

Check: insert flush; tray slides fully out and holds the laser without tipping.

Step 12 — Drawers, pull-outs and fronts

Drawing: [612-step-12.svg](#), Sheets 2-4.

Parts: all -DS, -DF, -DB, -FR drawer parts; BIT-, CR-; DOOR-L, DOOR-R; DOOR-BAY x2; LT-FR.

Tools: table saw with a 1/4" dado or a router table for the bottom grooves, pocket-hole jig or a lock-rabbit setup, drill, slide jig, chamfer bit, spacers for reveals.

1. Drawer boxes from 1/2" BB: sides, front and back, 1/4" bottom in a groove. Box width = opening minus 1"; depth per the cut list (36" for the SW and W1 drawers, 30" for W2).
2. Mount the slides at the heights on Sheets 2 and 4, boxes in.
3. Bit pull-out: a 12"-deep, 15"-wide, 17-1/2"-tall box with three drilled bit boards (1/4" and 1/2" holes on 1-1/2" centres, both faces), on 250 lb slides in the SE cabinet's east opening.
4. Clamp rack: two uprights notched for four 1-1/2" maple bars, base and top; on 250 lb slides in the SE cabinet's south opening. Bar spacing: 5", 10", 15", 20" from the bottom so F-clamps of every size hang without touching.
5. Fronts: cut each to its opening minus 1/8"; rout the 30-degree finger-pull chamfer along the top edge (3/4" deep); paint the faces charcoal and clear-coat the birch edges. Hang with 1/8" reveals using spacers; fix drawer fronts from inside the boxes with 4 screws each.

6. Router doors: full-overlay concealed hinges; the right door carries the paddle switch.
7. Flip-bay doors: two 16" doors on 110-degree hinges on the SW and SE sides; the right door carries the miter paddle switch and the flip checklist plate. They must open past 90 degrees so the swinging saw clears them.

Check: every front closes with even reveals; pull-outs run fully out and back with no rub.

Step 13 — Vise, dogs, lighting and finish

Drawing: 613-step-13.svg, Sheet 4.

Parts: vise, VISE-JAW liners, CRADLE x6, LED strip, dog-hole accessories.

Tools: drill, 3/4" Forstner, wrenches, finishing supplies.

1. Bolt the face vise to the pad at y 84-94 with its rear jaw flush with the west edge band; fit the maple liners; drill a 3/4" dog hole in the front jaw in line with the x = 4 column.
2. Make the six pipe-clamp cradles: a maple block with a 3/4" dowel peg and a saddle for 3/4" pipe; they drop into any dog hole and turn the field into a panel-glue-up station.
3. Stick the LED strip in the top's overhang on the W, S and E faces; driver on the main strip, PIR at the south face.
4. Finish: three coats of hardwax oil on the top, leaves, insert and maple; two coats of shellac inside the cabinets; charcoal enamel on the exposed base sides.
5. Vacuum every cavity through the gates; run each station; write the VERIFY measurements you took inside the utility cover lid for the next person.

Check: vise closes square; a holdfast set in any hole holds a plane stroke; LED on when you walk up; all gates labelled.

Dust collection

Dust collection

Drawings: Sheet 8 ([renders/32-section-C-ducts.svg](#)), Sheet 11 ([renders/50-dust-routing.svg](#)), step view [605-step-05.svg](#).

Two systems, one chase

You have a Harbor Freight collector with 4" ducting on this side of the shop and a DeWalt shop vac behind a cyclone separator. They do different jobs and want different pipe:

	HF collector	Shop vac + cyclone
Moves	high volume, low pressure (~600-800 CFM claimed, 350-450 real through 4")	low volume, high pressure (~100-150 CFM through 2-1/2")
Good at	hoods, chips, the table saw, the router box, the floor sweep, the downdraft field	small ports right at the cut: router fence, miter saw chute
Pipe	4"	2-1/2"

So the bench carries **both trunks side by side** in the 5" x 8" spine chase at floor level: the 4" main below, the 2-1/2" vac trunk above it. Each has its own port on the east face at the utility column (4" at 6" above the floor, 2-1/2" at 10"), so you connect each machine to its own trunk and never need a reducer.

Drops

#	Drop	Trunk	Gate location	Notes
1	Floor sweep	4"	south toe kick, behind a flap	sweep the shop floor straight into it

2	Miter hood	4"	flip-bay east wall	4' of 4" flex with a quick-connect cuff; plug it onto the hood after each flip
3	Router box	4"	router-box floor	box is sealed; 1" x 4" make-up-air slot on the north partition
4	Downdraft plenum	4"	service gap (reach through the W-BAND1 clean-out)	81 holes x 0.44 sq in = 36 sq in of open area; at 350 CFM that is ~1,400 fpm through the holes, plenty for sanding
5	Table saw	4"	motor-bay south wall	4' of 4" flex to the CNS's own port
6	Router fence	2-1/2"	behind the flip-lid port in the east edge band	your fence's hose plugs into the face of the bench
7	Miter saw chute	2-1/2"	flip-bay east wall	3' of 2-1/2" flex to the DWS779's port, connected after the flip

Rules that make it pull

1. **One 4" gate open at a time** on the HF collector. It cannot feed two hoods at once through 4" pipe.
2. **Wyes, not tees.** Every branch enters the trunk through a 45-degree wye pointing toward the port.
3. **No 90s.** Two 45s with a short straight between them where the run must turn (the leg at the north end).
4. **Straight and short.** The spine is 79" of straight pipe. The longest drop (miter hood) is 6' of flex; everything else is under 3'.
5. **Seal the boxes.** Silicone every plenum joint; foam-tape the router doors; the router box has one deliberate air inlet and nothing else.
6. **Clean-outs.** The floor-sweep gate doubles as the south clean-out; the utility-column flanges are the north clean-out.

7. **Static.** Ground the 4" run with a bare copper wire inside if you use PVC and get shocks; not strictly needed on a run this short.

The miter hood

A 28" wide x 6" deep x 14" tall box on the flip core behind the saw, open toward the blade, with a floor sloped to a 4" port low at the back. Height 14" puts its top about 9-1/2" above the bench top when the saw is up. It flips with the saw; the hose has a quick-connect cuff that you push on

after the platform is locked. Add a strip of 1/8" brush or rubber flap across the top edge if you want to catch the high spray from a slider.

Optional automation

An iVac-style tool sensor on the router and miter outlets can start the collector and open a motorised gate. The wiring is on the main strip; the gates replace the manual ones at drops 2 and 3. Do this later if the manual routine bothers you.

Electrical

Electrical

Drawings: Sheet 12 (renders/51-electrical.svg), step view 606-step-06.svg .

Everything is 120 V on a single 20 A circuit, exactly like your other bench, built from off-the-shelf parts and no hard wiring inside the bench. If you later bring a second 20 A circuit, put the router on it.

Feed

- A flush-mount 20 A inlet (twist-lock L5-20 or a straight-blade 5-20) in the utility cover on the east face, 18" above the floor.
- A 12 AWG SJOOW cord from the inlet to the wall or a ceiling drop. It is the only cord that leaves the bench.

Distribution

- **Main strip:** a metal 12-outlet, 15 A surge strip with a lit switch mounted inside the utility column. Its switch is the bench master.
- **Raceway:** 1" surface raceway on top of the spine lid (never inside the dust chase) to the south end, and through the service gap to the west cabinets.
- **Branch strips:** metal 6-outlet strips inside W1 (charging drawer, LED driver), in the flip bay (miter), and behind the east face (router box, east USB strip).

Outlets and switches

Where	What	Notes
Router box (inside, high on the north partition)	single outlet	router plugs in here; cord through a grommet

Right-hand router door front	paddle safety switch	switches the router outlet
Flip-bay east wall, 12" up	single outlet through the seat switch	miter saw; plug in after the platform is seated; no power unless seated and clamped
Right-hand flip-bay door	paddle safety switch	switches the miter outlet (in series with the seat switch)
W1 top drawer	switched outlet in the drawer back	charging drawer; 1" vent slots front and back
W2 (inside, north wall)	single outlet	laser
West face, y 50-58, under the overhang	flush strip, 2 outlets + USB-A/C	hand tools, phone
South face, x 56-64	flush strip, 2 outlets + USB-A/C	miter side
East face, y 68-74	flush strip, 2 outlets + USB-A/C	router side
North face beside the motor bay	single outlet	the table saw plugs in here

Loads

The router (DW618, 12 A) and the DWS779 (15 A) each take most of the circuit. Run one at a time. The collector and the vac are on their own circuits at the wall. USB strips, LEDs and chargers are negligible.

Lighting

12 V warm-white LED strip in the top's overhang on the west, south and east faces, driver on the main strip, PIR sensor at the south face so it lights when you walk up. The light grazes the fronts and reads as a floating top.

Safety

- Metal strips and boxes only; screw them to plywood, never to the duct.
- Keep every cord out of the dust chase; the raceway sits on the lid.
- Label each plug at the main strip.
- The master switch and the two paddles are the kill switches; nothing is wired into the mains.

Using the stations

How each station works

Outfeed (north)

Roll the saw against the bench so its rear-rail face touches the bridge lip. The top sits 1/16" below the saw table so stock never catches; the two grooves take a miter gauge bar or a sled runner for 14". Anything ripped travels the full 96" over a flat top because the miter saw is below it. The motor bay behind the north face gives the CNS motor 15" of room and hides the saw's 4" hose.

Hand-work and clamping station (west)

- **Vise:** 9" quick-release at the NW corner, maple-lined, dog in the front jaw in line with the first column of holes.
- **Dog field:** 81 holes, 4" grid, 2-1/4" of material. Rockler dogs, holdfasts, hold-downs and surface clamps all work.
- **Clamping station:** drop the six pipe-clamp cradles into the holes at the spacing you need and lay the pipe clamps in them; panels glue up flat and square. Cauls clamp to the dog holes. The 1" overhang on the west and south edges takes F-clamps anywhere.
- **Downdraft:** open the plenum gate (service-gap side, reach through the band-panel clean-out) and the whole field pulls sanding dust down. Close it when clamping so you are not fighting the vacuum.
- **Drawers:** W1 has three drawers (top one is the switched charging drawer); W2 has the laser tray over a deep drawer.

Router station (east)

- Raise the bit with the Rout-R-Lift crank through the plate; the plate is flush when the bit is retracted, so the top stays a bench.

- Clamp your fence into the two T-tracks 26" apart; slide it to the bit. Plug the fence hose into the flip-lid port in the edge band.
- Featherboards and a miter gauge use the combo track 4-1/2" from the edge.
- Bits live in the vertical pull-out beside the doors: pull it out, take the bit, push it back. Collets and wrenches on its top shelf.
- The router box is sealed and ducted; open its gate on the spine (inside the box floor) before routing.

Miter station (south)

The saw lives on a two-sided platform that pivots on a steel axle inside the base. One face carries the DWS779, the other a hollow birch-faced box the same height as the saw's deck, so whichever side is up sits flush with the top.

To bring the saw up

1. Open both bay doors fully. Check the 30" in front of the bench is clear.
2. Release the two toggle clamps, then pull the two index plungers.
3. With a hand on each side of the platform, rotate it half a turn: the saw comes up through the hatch as the box goes down. It is balanced by the steel in the box, so it moves at walking pace and stops where you leave it.
4. Let the platform settle onto the four rest blocks; the plungers click in on their own. Close the toggle clamps.
5. Push the 4" cuff onto the hood and the 2-1/2" hose onto the saw's chute; plug the saw in. The seat switch will not pass power until the platform is seated and clamped.
6. Raise the drop-leaves on both ends (the brackets lock). Open the miter-hood and chute gates. Cut.

To stow it

1. Switch off at the paddle. Lock the saw's head down, lock the slide, lock the bevel, close the guard. Unplug and pull the two hoses.
2. Drop the wings. Release the clamps, pull the plungers, rotate the platform back: the saw swings out through the open doors and under; the box comes up flush. Plungers click, clamps on.
3. Close the doors. The top is flat.

What keeps it safe: the load sits on the rest blocks, never on the pins; the plungers locate; the toggle clamps hold it down against vibration; the seat switch means the saw cannot run unless all of that is true; the counterweight means a released platform does not fall. Do not flip with the head unlocked.

Long stock: the north half of the bench takes anything up to 8' to the left or right when you angle it across the top; for 12' stock use a roller stand beyond a drop-leaf.

Laser well (top, north-west)

- Flat work: leave the insert in; the laser sits on the top like any desk.
- Rotary work: lift the insert out (finger hole), drop the floor panel onto the cleats at the depth you need (2", 4-1/2" or 7"), set the laser over

the opening so the rotary attachment hangs inside. Plug into the W2 outlet; run the exhaust hose to a window.

- Storage: pull the tray out of W2, set the laser on it, push it home.

Storage map

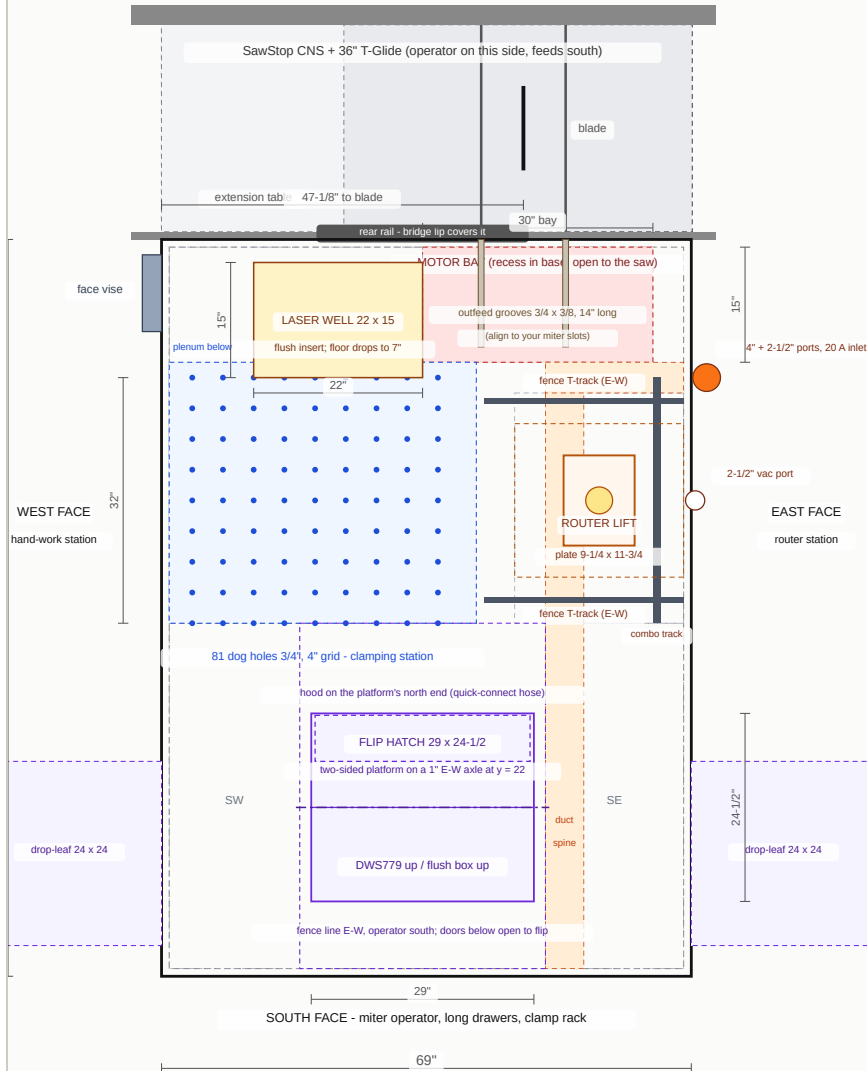
Face	Opening	Contents
South	SW-D1/D2/D3, 13-1/2" x 36" long drawers	straightedges, levels, your router fence, track-saw rail, long clamps
South	flip-bay doors	the platform, saw stowed underneath; flip checklist on the door
South	clamp rack pull-out	F-clamps and bar clamps on four bars
East	bit pull-out	router bits, collets, wrenches
East	router doors	the lift and router; push blocks and featherboards on the side compartments
East	utility column	inlet, main strip, both dust ports
West	W1 drawers	charging drawer; marking and measuring; chisels and planes
West	W2	laser on its tray; deep drawer for sanding supplies and dog-hole accessories

Dimensioned sheets

Plan, four elevations, three sections, station details, dust and electrical routing. Sheet numbers match the build guide.

SHEET 1 - PLAN VIEW (top surface) with base layout below, dashed

Island 69 x 96 in. North (top of page) is the table-saw side. Blade line, miter slots and outfeed grooves are aligned to the SawStop CNS with the 36" extension on the west.

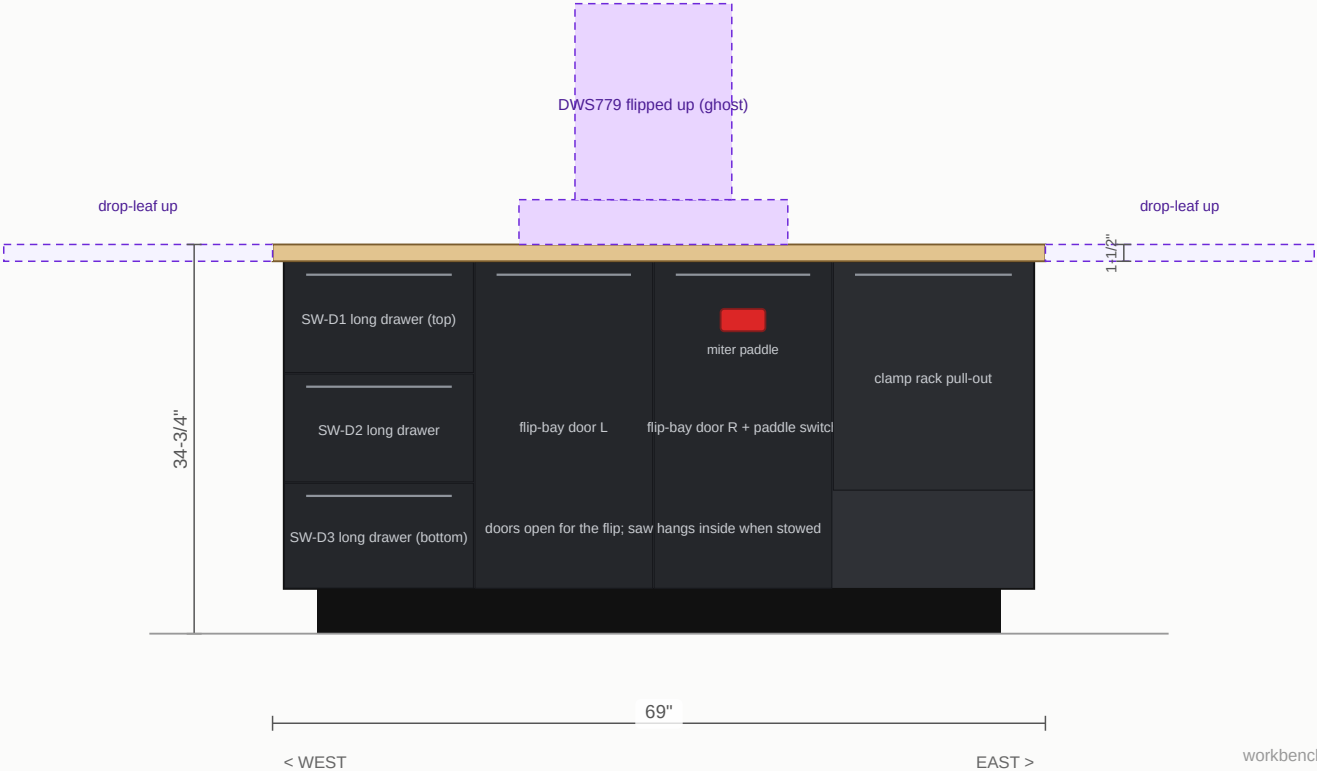


workbench-plans | generated from tools/model.py | inches

20-plan.svg

SHEET 2 - SOUTH ELEVATION (miter station face)

Base is 33-1/4 tall on a 4" recessed plinth; top is 1-1/2 thick with 1-1/2 maple edge. Fronts: 3/4 Baltic birch, painted charcoal, routed finger pulls.

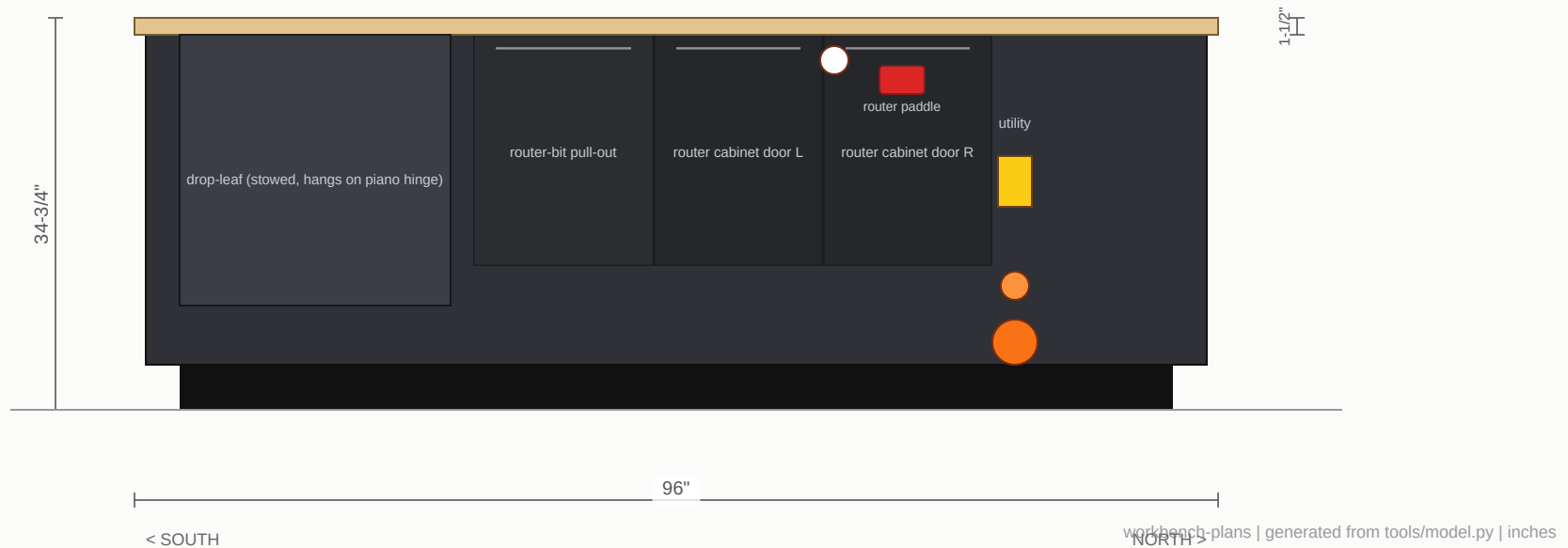


workbench-plans | generated from tools/model.py | inches

21-elevation-S.svg

SHEET 3 - EAST ELEVATION (router station face)

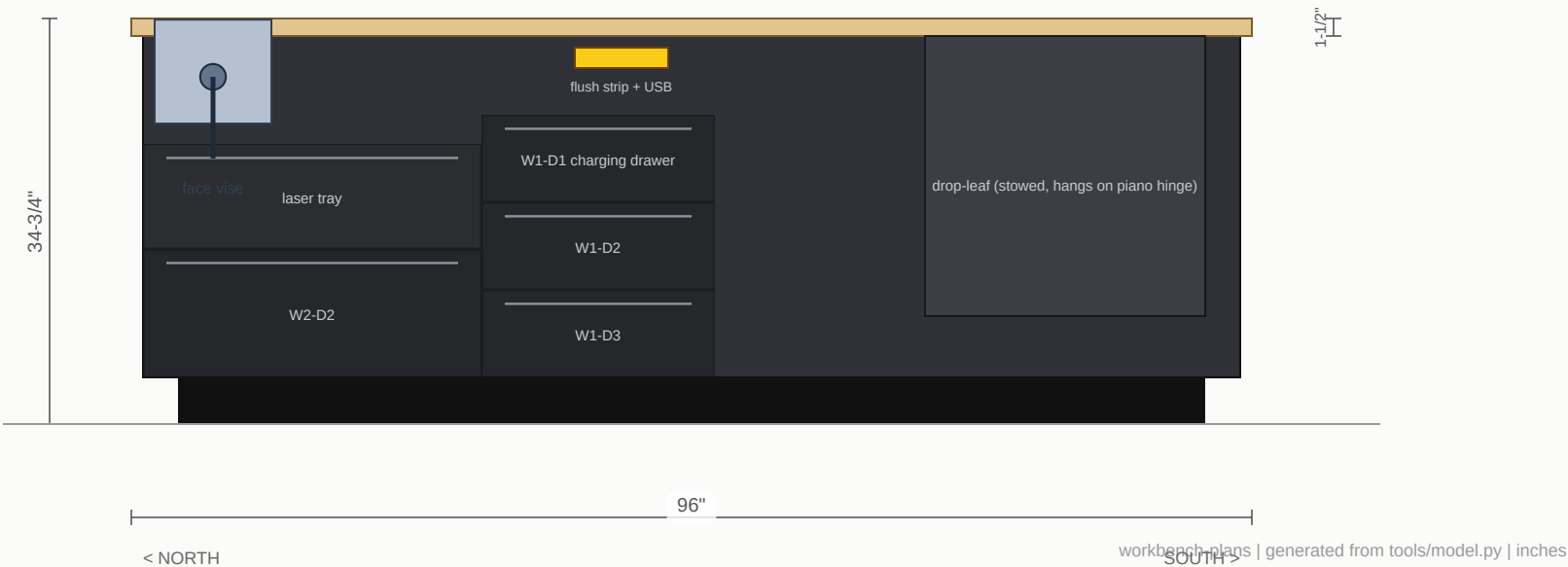
Base is 33-1/4" tall on a 4" recessed plinth; top is 1-1/2" thick with 1-1/2" maple edge. Fronts: 3/4" Baltic birch, painted charcoal, routed finger pulls.



22-elevation-E.svg

SHEET 4 - WEST ELEVATION (hand-work face)

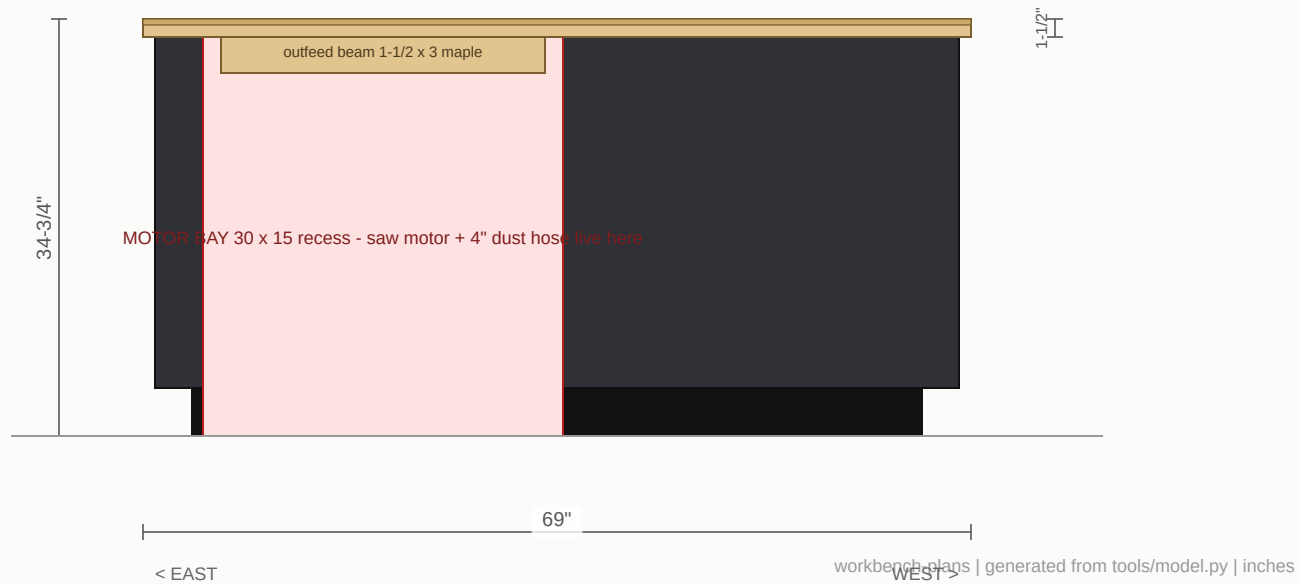
Base is 33-1/4" tall on a 4" recessed plinth; top is 1-1/2" thick with 1-1/2" maple edge. Fronts: 3/4" Baltic birch, painted charcoal, routed finger pulls.



23-elevation-W.svg

SHEET 5 - NORTH ELEVATION (outfeed / table-saw side)

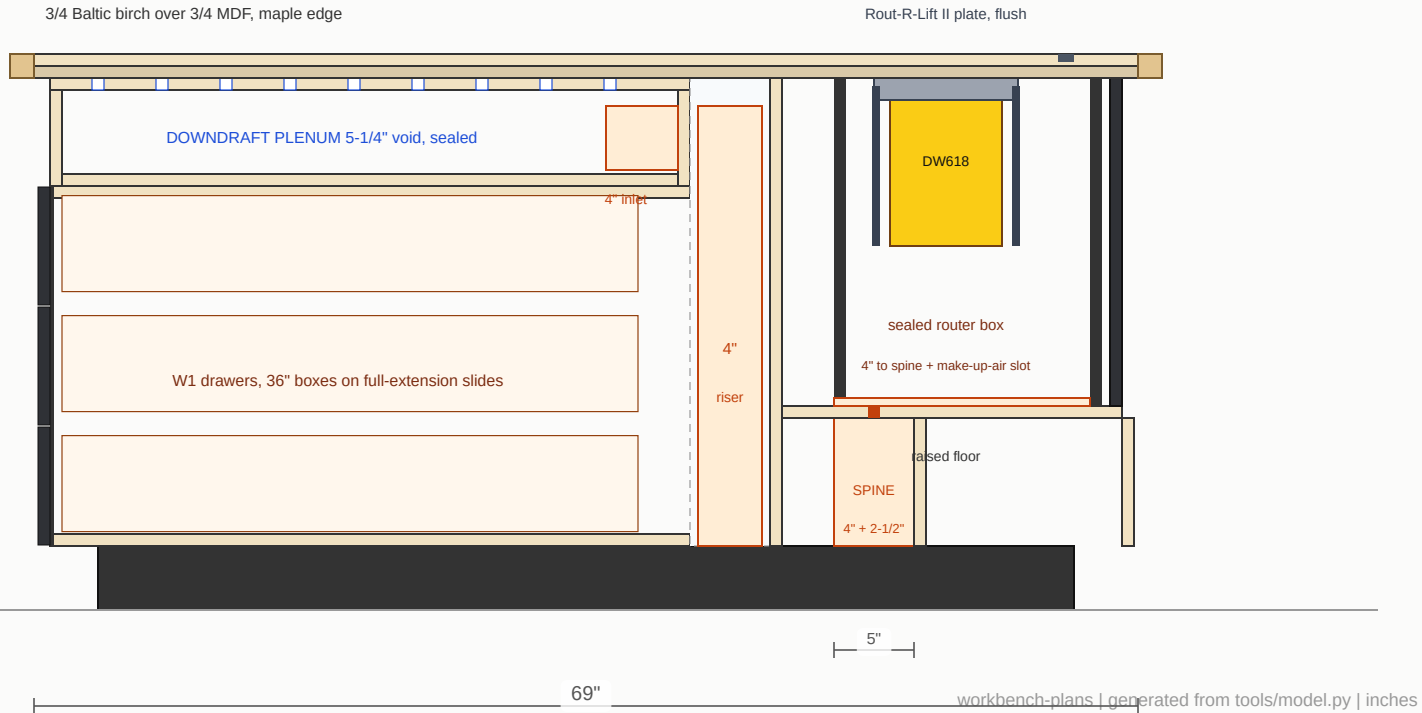
Base is 33-1/4" tall on a 4" recessed plinth; top is 1-1/2" thick with 1-1/2" maple edge. Fronts: 3/4" Baltic birch, painted charcoal, routed finger



24-elevation-N.svg

SHEET 6 - SECTION A-A (east-west at y = 62, through the router lift; looking north)

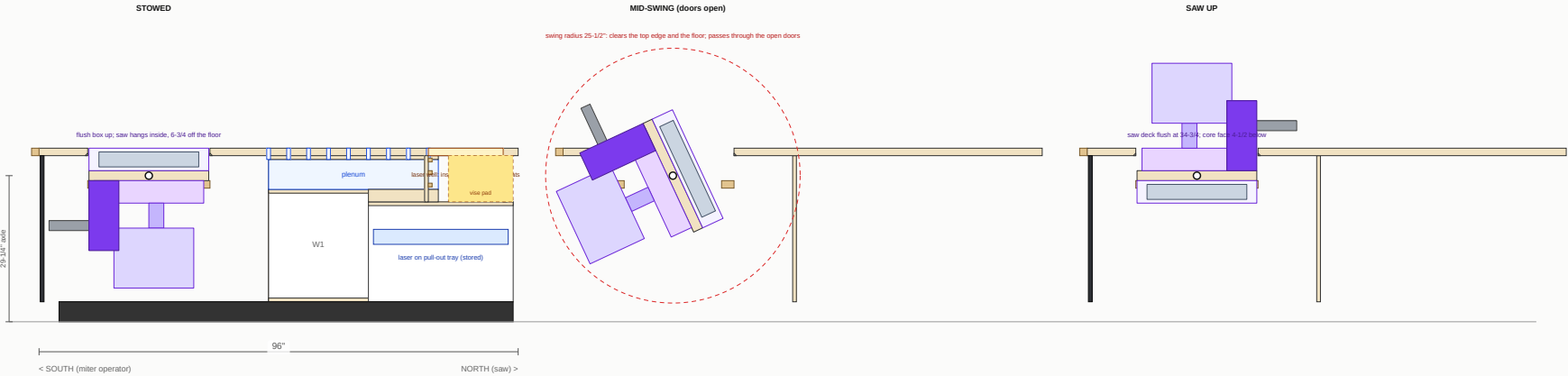
Shows the top laminate, downdraft plenum, W1 drawers, service gap with the plenum riser, spine duct under the router cabinet, and the sealed router box.



30-section-A-router.svg

SHEET 7 - SECTION B-B (north-south at x = 30, through the flip bay, plenum, W2 and laser well; looking east)

Left: saw stowed, flush box up, top flat. Middle: mid-swing with the doors open, saw passing through the front. Right: saw up, seated on the rest blocks. Same axle, same platform.

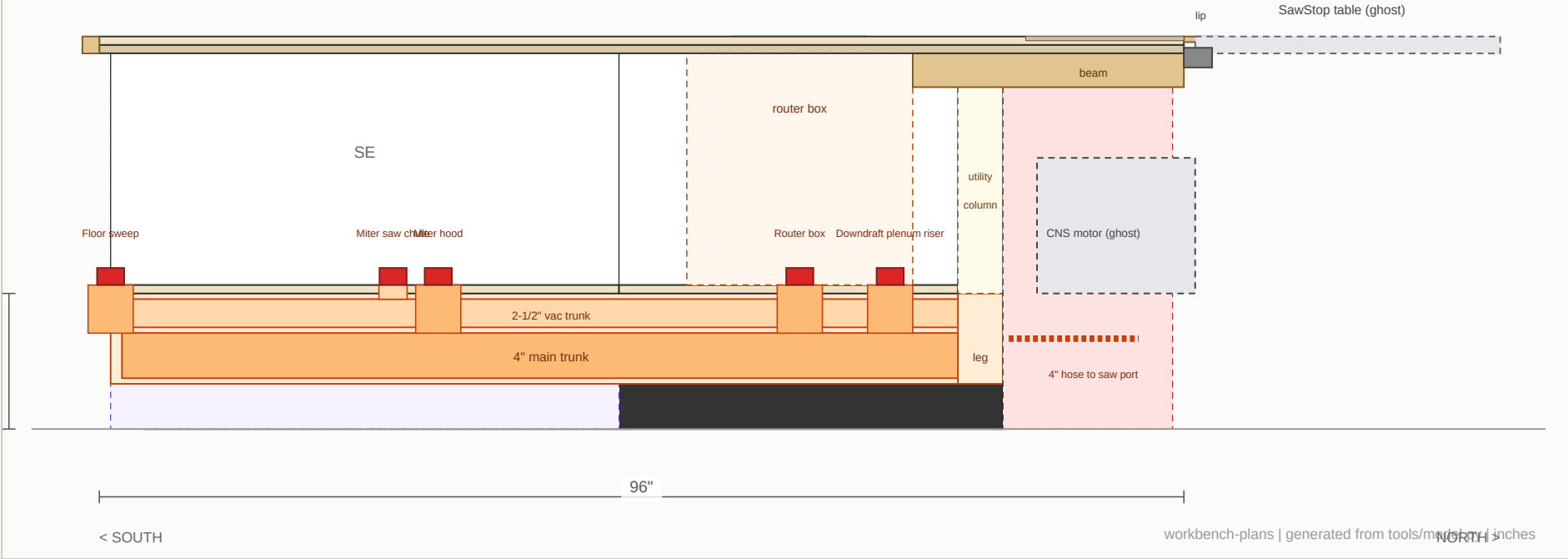


workbench-plans | generated from tools/model.py | inches

31-section-B-lift.svg

SHEET 8 - SECTION C-C (north-south at x = 52, through the spine duct, SE cabinet, router cabinet, duct leg and motor bay; lo

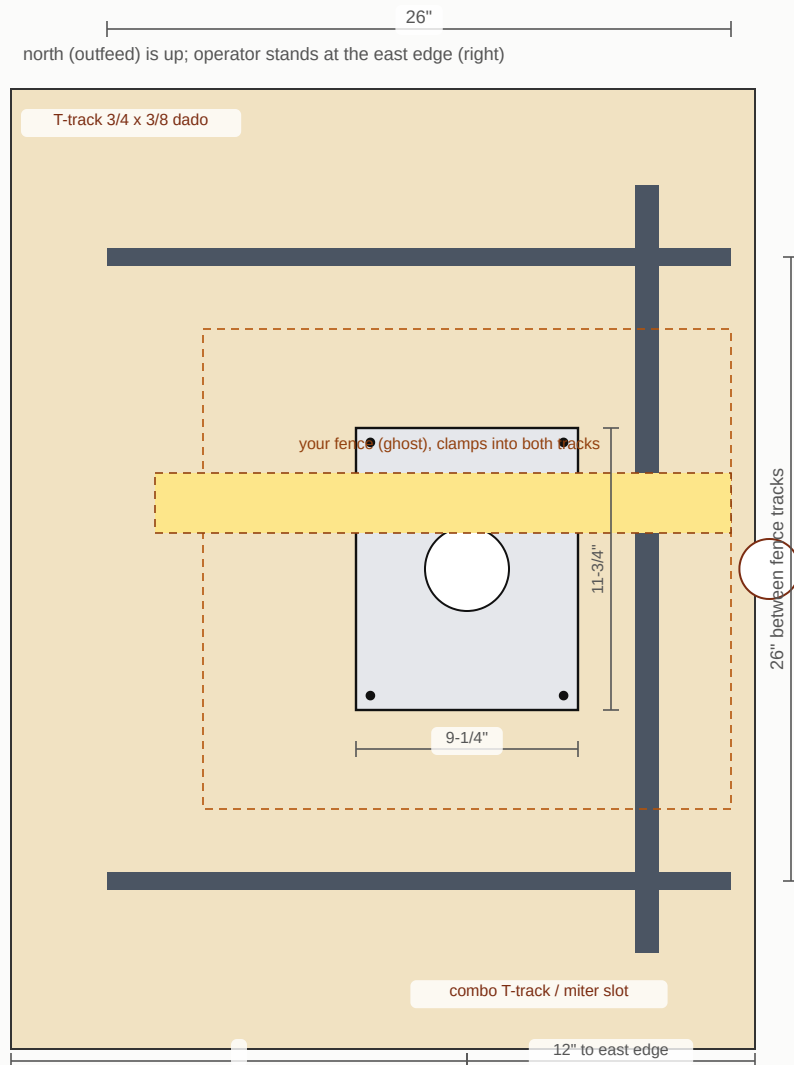
The 4" main and 2-1/2" vac trunks run side by side in the spine at floor level. Every drop is a 45-degree wye with a blast gate. The table saw hose exits into the motor bay.



32-section-C-ducts.svg

SHEET 9 - ROUTER STATION DETAIL (plan, east half of the top)

Plate recess for a JessEm Rout-R-Lift II ($9\text{-}1/4 \times 11\text{-}3/4 \times 3/8$, leveling screws in the corners). Fence tracks are 26" apart; check your fence's clamp spacing before cutting the dados.

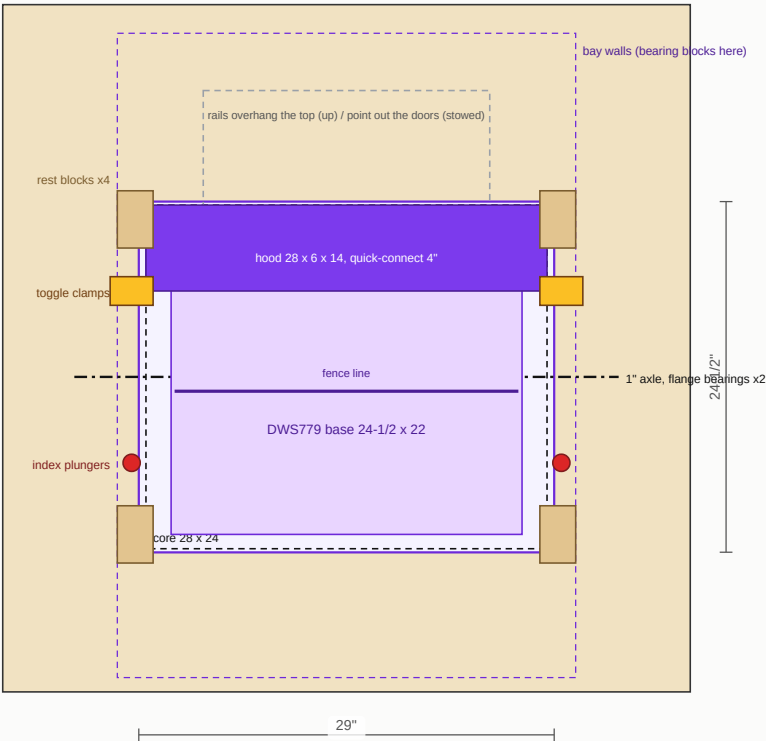


workbench-plans | generated from tools/model.py | inches

40-router-station.svg

SHEET 10 - FLIP-TOP PLATFORM (plan) and the safety system

Two-sided platform: 2" core (two 3/4 BB skins on 1/2 BB ribs) with the saw bolted to one face and a 4-1/2" hollow flush box on the other. A 1" steel axle in flange bearings on the bay walls. Both faces sit the same distance from the axle, so e



How it stays safe

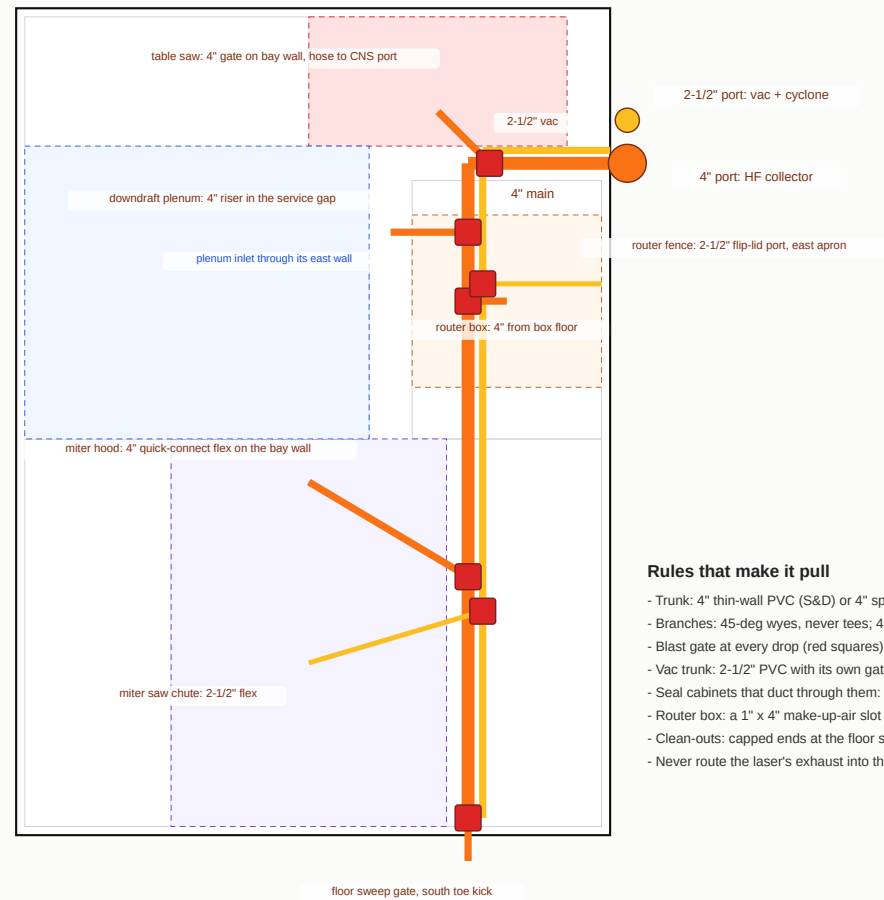
- Load path** The platform seats on four maple rest blocks (both ways up); the axle only locates it.
- Locate** Two 1/2" spring index plungers snap into the bay walls at 0 and 180 deg.
- Clamp** Two 500 lb hold-down toggle clamps pull the core onto the rest blocks. No rattle, no lift-off.
- Interlock** A plunger seat switch under one rest block is in series with the paddle: no power unless seated and clamped.
- Balance** ~70 lb of steel plate in the flush box brings the saw side to within ~13 ft-lb of balance.
- Swing** Doors open first; the saw swings out through the front. Keep 30" clear. Two hands on the platform edges.
- Saw** Head locked down, slide locked, bevel locked, blade guard closed before every flip. Checklist plate on the door.
- Axle** 1" cold-rolled rod, two 1" flange bearings on 1-1/2" doubler blocks, shaft collars with set screws.
- Hose / cord** 4" quick-connect cuff and a plug on the bay wall; connect after the plungers click in.

Verify before building: DWS779 locked-down height (21-1/2 assumed), deck height (4-1/2 assumed), base depth (22 assumed).

SHEET 11 - DUST COLLECTION ROUTING (plan)

One 4" main trunk for the Harbor Freight collector and one 2-1/2" trunk for the shop vac + cyclone, side by side in the spine. Wyes point toward the ports; blast gate at every drop; cabinets sealed.

north (table saw) is up



Rules that make it pull

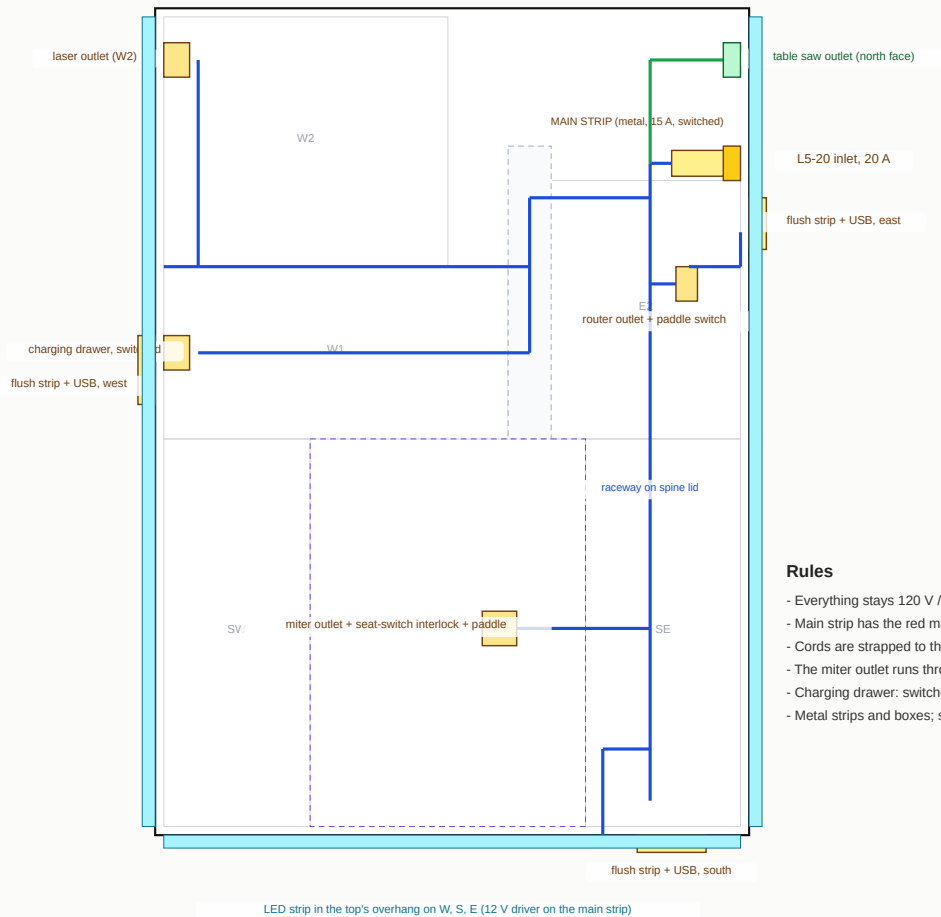
- Trunk: 4" thin-wall PVC (S&D) or 4" spiral, straight, floor level.
- Branches: 45-deg wyes, never tees; 45-deg elbows in pairs instead of a 90.
- Blast gate at every drop (red squares). Open ONE 4" gate at a time on the HF collector.
- Vac trunk: 2-1/2" PVC with its own gates; the cyclone sits before the vac.
- Seal cabinets that duct through them: silicone the plenum, foam tape on doors.
- Router box: a 1" x 4" make-up-air slot near the door so it does not starve.
- Clean-outs: capped ends at the floor sweep and the utility corner.
- Never route the laser's exhaust into this system (fire risk).

workbench-plans | generated from tools/model.py | inches

50-dust-routing.svg

SHEET 12 - ELECTRICAL LAYOUT (plan)

One 20 A 120 V feed by 12 AWG SJOOW cord to a flush L5-20 inlet in the utility column. Distribution by a metal 15 A surge strip; branch strips inside cabinets; flush strips with USB-A/C on the faces.



Rules

- Everything stays 120 V / 20 A. Never run the router and the miter saw at the same time (each ~15 A)
- Main strip has the red master switch; paddle switches at router and miter are the tool switches.
- Cords are strapped to the raceway, never loose in the dust chase; use 12 AWG for the feed, 14 AWG for the branch.
- The miter outlet runs through the seat switch: no power unless the platform is seated and clamped.
- Charging drawer: switched outlet, 1" vent slots front and back.
- Metal strips and boxes; screw the strips to plywood, not to the duct.

workbench-plans | generated from tools/model.py | inches

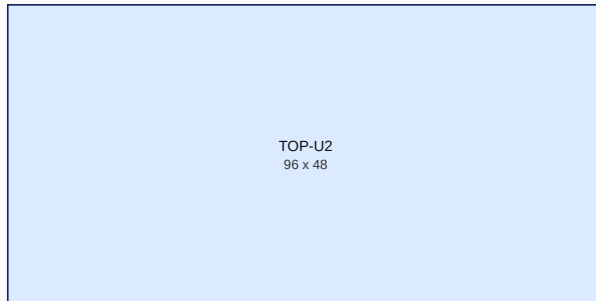
51-electrical.svg

Cut diagrams

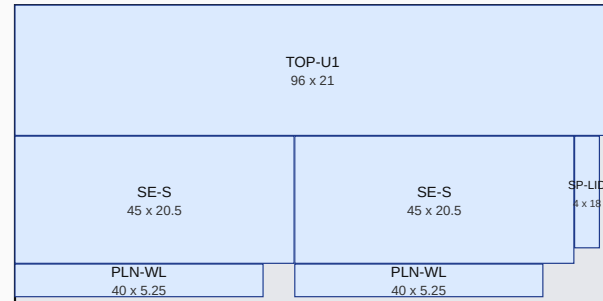
Nested 4x8 layouts with 1/8" kerf. Part IDs match the cut list. Grey is offcut.

CUT DIAGRAM - 3/4" Baltic birch (4x8) - sheets 1 to 6 of 12

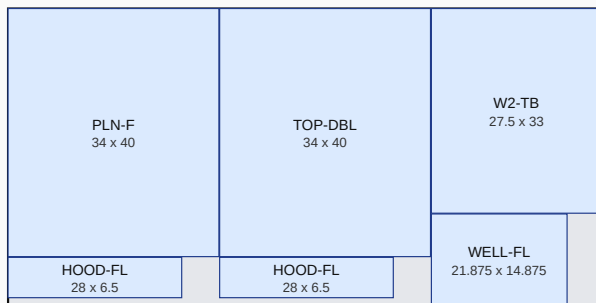
12 sheets of 3/4" Baltic birch (4x8) in total. 1/8" kerf allowed. Part IDs match docs/03-cut-list.md. Grey = offcut.



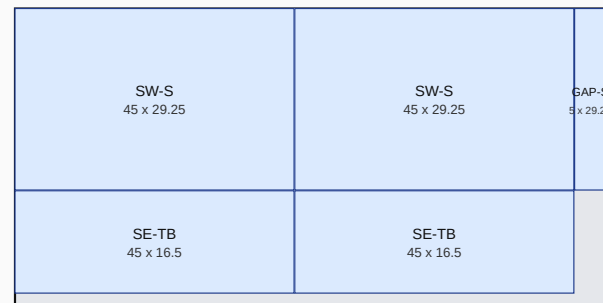
Sheet 1 100% used



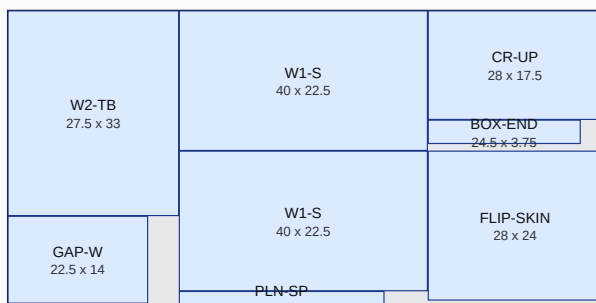
Sheet 2 94% used



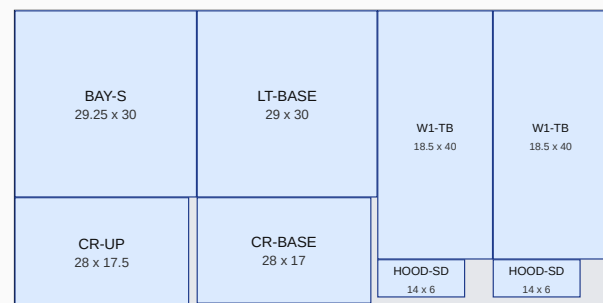
Sheet 3 94% used



Sheet 4 93% used



Sheet 5 95% used

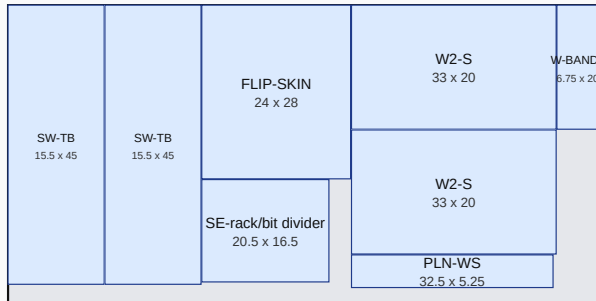


Sheet 6 95% used

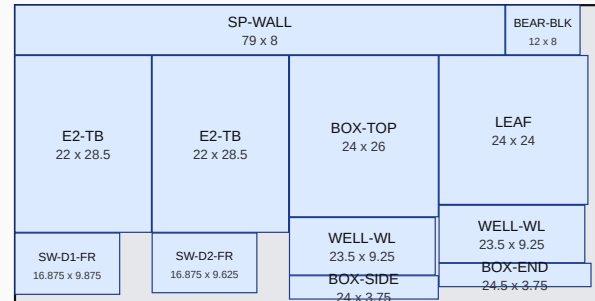
cuts/BB34-sheets-1.svg

CUT DIAGRAM - 3/4" Baltic birch (4x8) - sheets 7 to 12 of 12

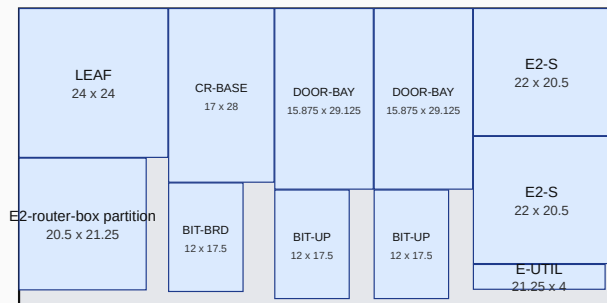
12 sheets of 3/4" Baltic birch (4x8) in total. 1/8" kerf allowed. Part IDs match docs/03-cut-list.md. Grey = offcut.



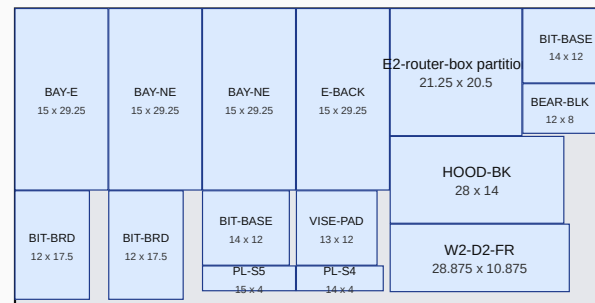
Sheet 1 87% used



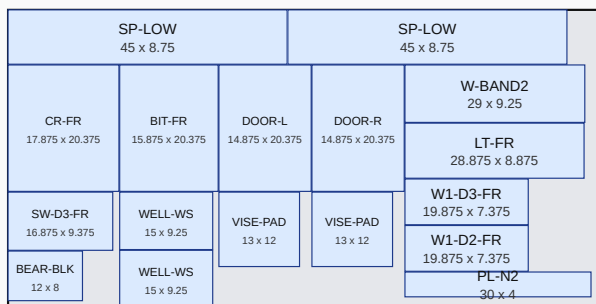
Sheet 2 90% used



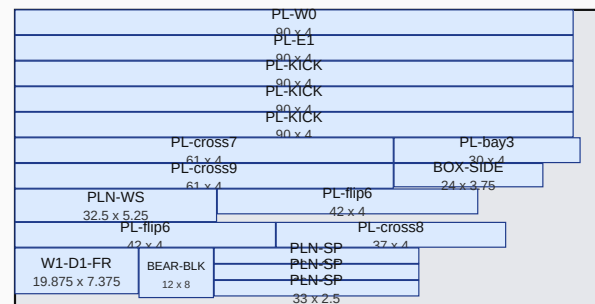
Sheet 3 87% used



Sheet 4 87% used



Sheet 5 84% used

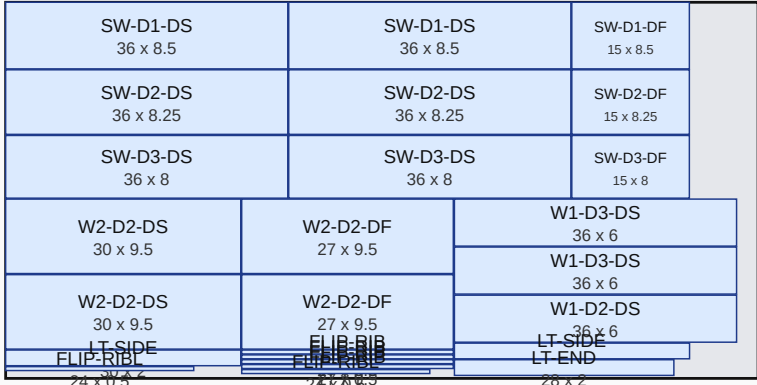


Sheet 6 79% used

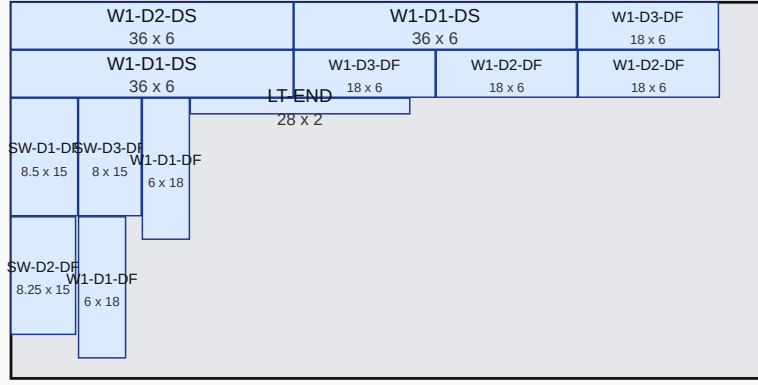
cuts/BB34-sheets-2.svg

CUT DIAGRAM - 1/2" Baltic birch (4x8) - sheets 1 to 2 of 2

2 sheets of 1/2" Baltic birch (4x8) in total. 1/8" kerf allowed. Part IDs match docs/03-cut-list.md. Grey = offcut.



Sheet 1 90% used

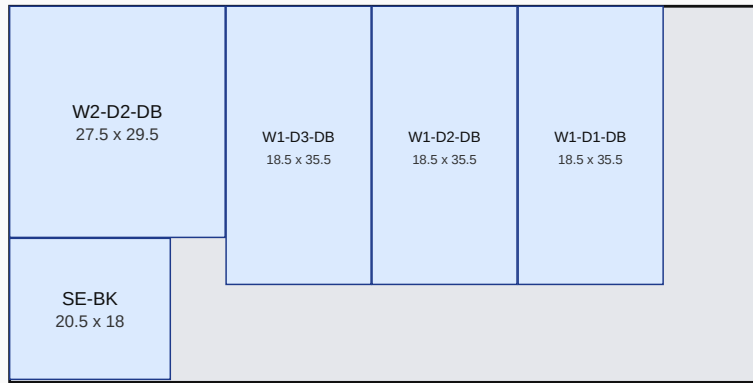


Sheet 2 37% used

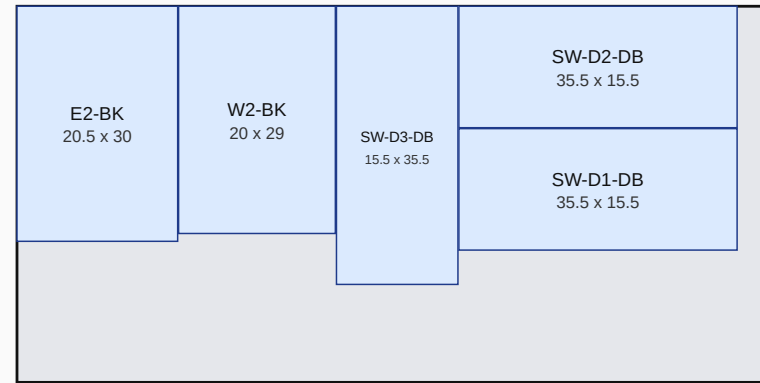
cuts/BB12-sheets-1.svg

CUT DIAGRAM - 1/4" Baltic birch (4x8) - sheets 1 to 3 of 3

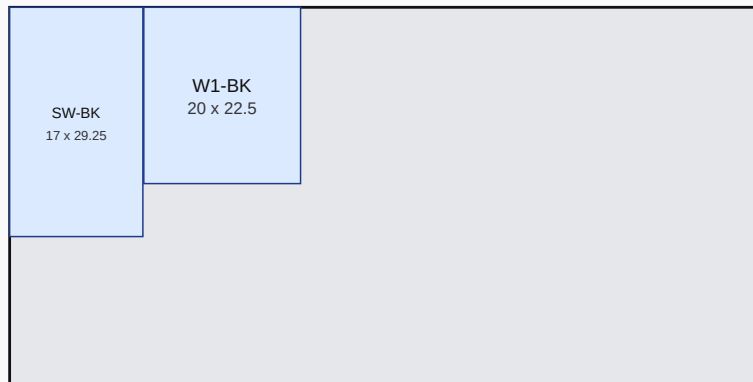
3 sheets of 1/4" Baltic birch (4x8) in total. 1/8" kerf allowed. Part IDs match docs/03-cut-list.md. Grey = offcut.



Sheet 1 68% used



Sheet 2 62% used

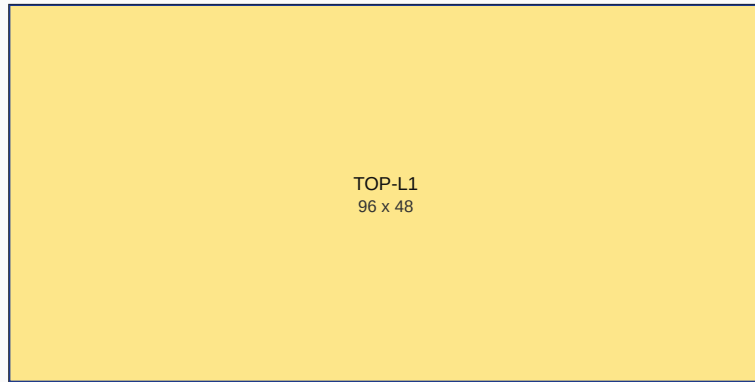


Sheet 3 21% used

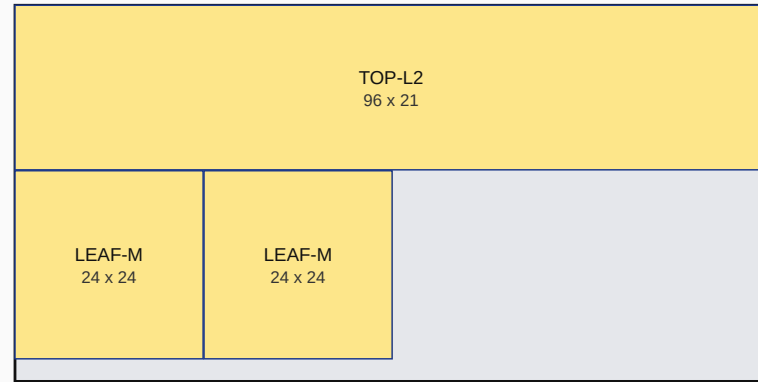
cuts/BB14-sheets-1.svg

CUT DIAGRAM - 3/4" MDF (4x8) - sheets 1 to 2 of 2

2 sheets of 3/4" MDF (4x8) in total. 1/8" kerf allowed. Part IDs match docs/03-cut-list.md. Grey = offcut.



Sheet 1 100% used

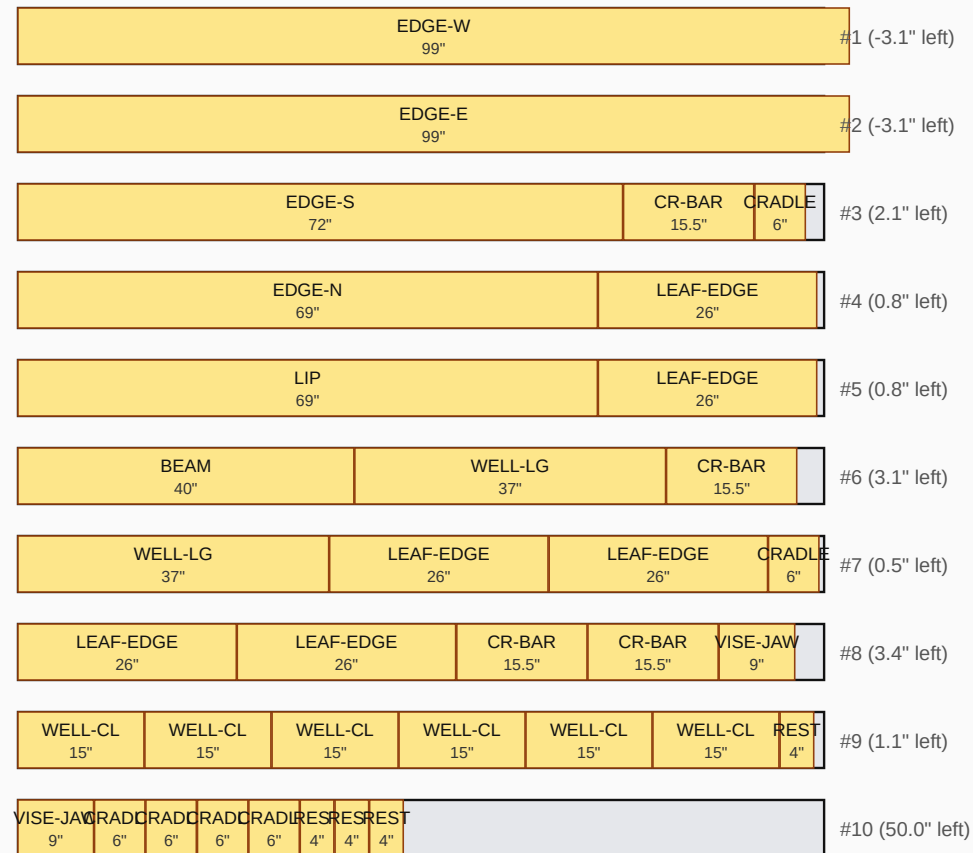


Sheet 2 69% used

cuts/MDF34-sheets-1.svg

CUT DIAGRAM - hard maple, 8/4 x 3" x 96" boards (rip to width after cross-cut)

10 boards of 8/4 hard maple, 3" wide x 8'. Rip each piece to its listed width; the 1-1/2" edging comes from the same stock.



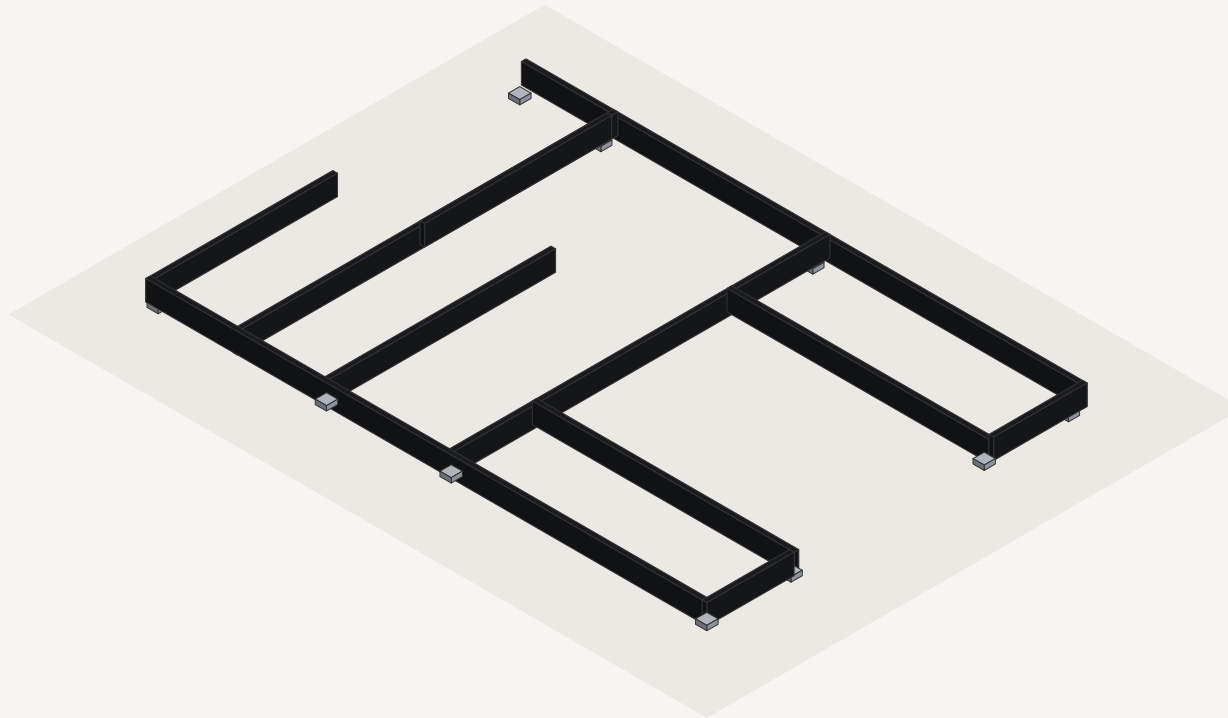
cuts/MAPLE-boards.svg

Build steps

One isometric view per build step; the step's new parts are in colour, earlier work is grey.

BUILD STEP 1 - Plinth and levelling feet

Ladder frame of 4" x 3/4" Baltic-birch ribs on ten 1,000 lb levelling feet; open under the flip bay (front centre). Recessed 3" on the S, E, W faces.

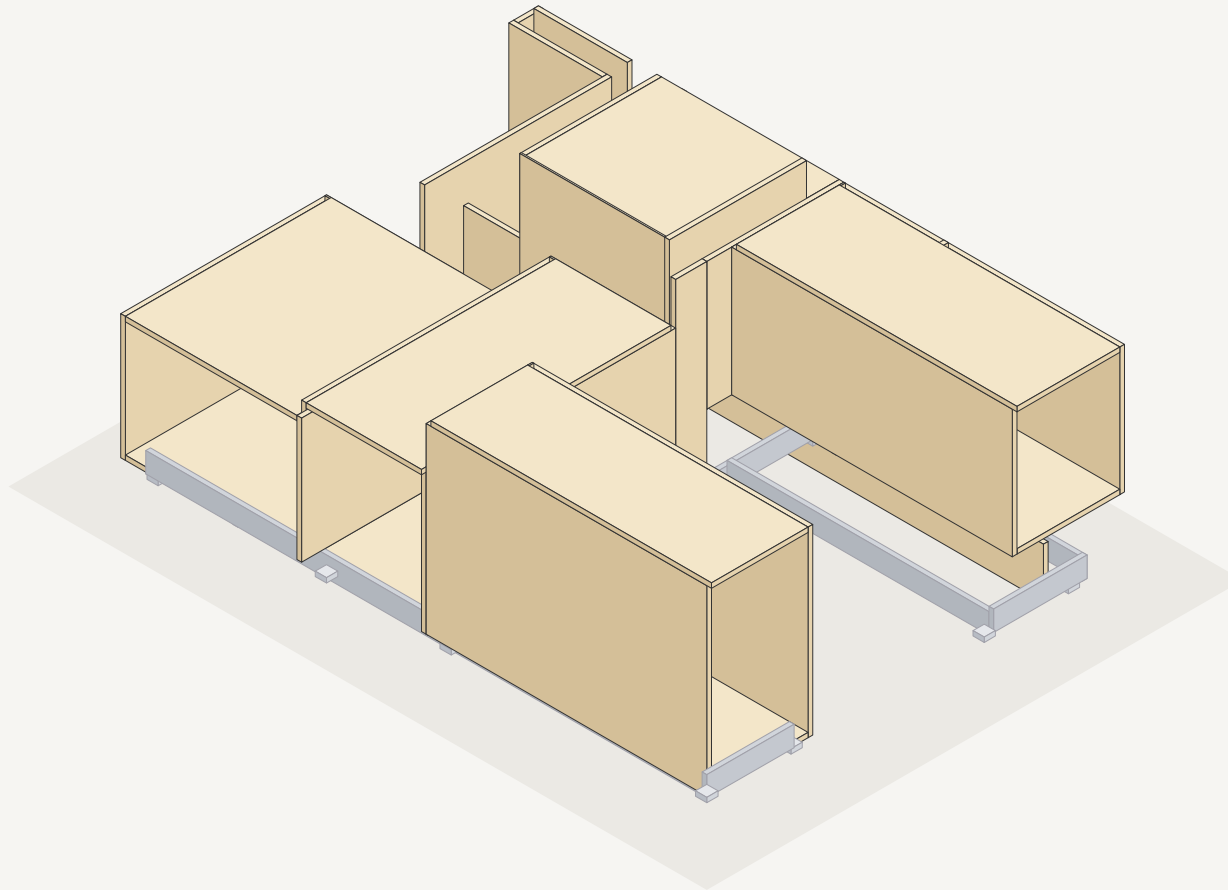


View from the south-west. Parts added in this step are in colour; earlier work is grey.

601-step-01.svg

BUILD STEP 2 - Carcasses (base cabinets, spine, bays)

Five frameless carcasses (SW, SE, W1, W2, E2), the spine duct wall, motor-bay walls, service-gap walls and router-box partitions. SE and E2 have raised floors over the spine.

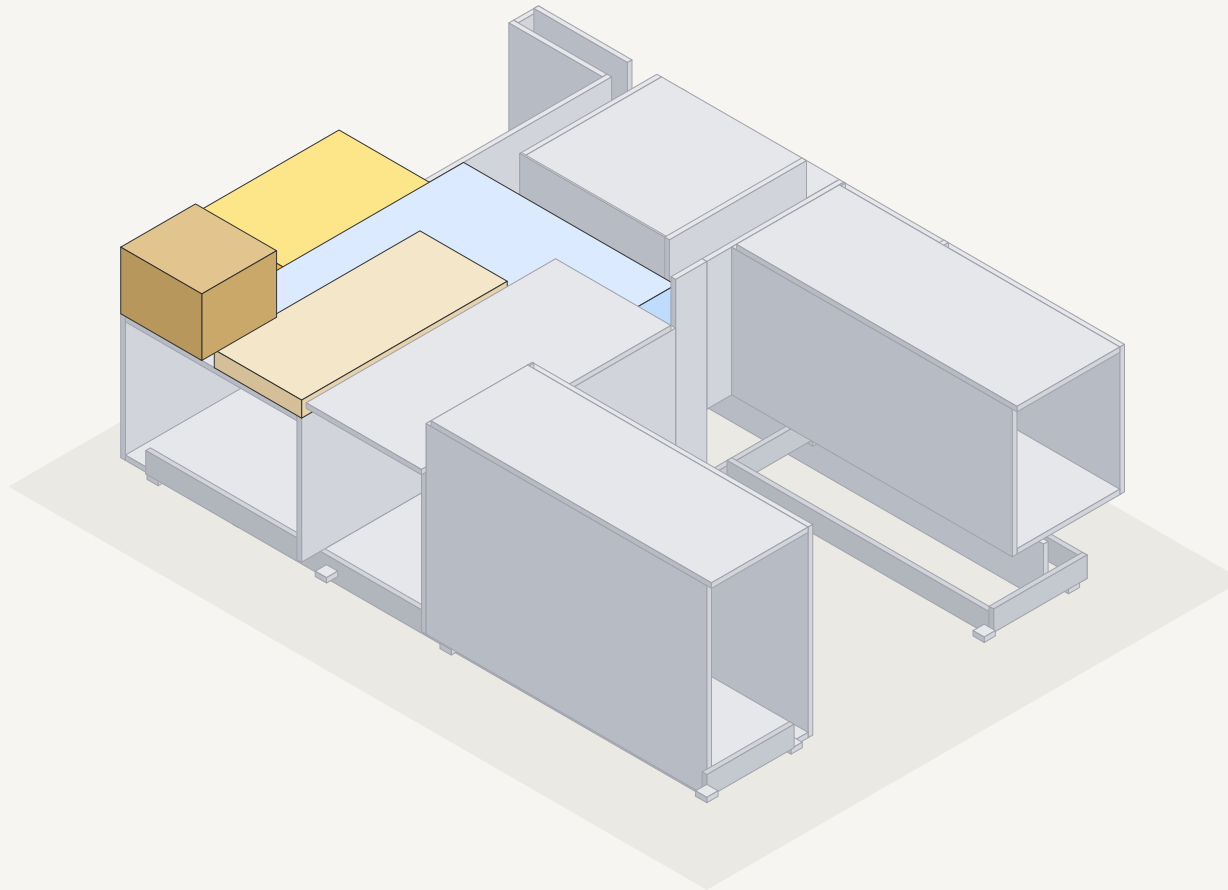


View from the south-west. Parts added in this step are in colour; earlier work is grey.

602-step-02.svg

BUILD STEP 3 - Downdraft plenum, laser well and vise block

Sealed downdraft plenum on W1/W2 (blue), spacer frame, laser-well box and the laminated vise pad.

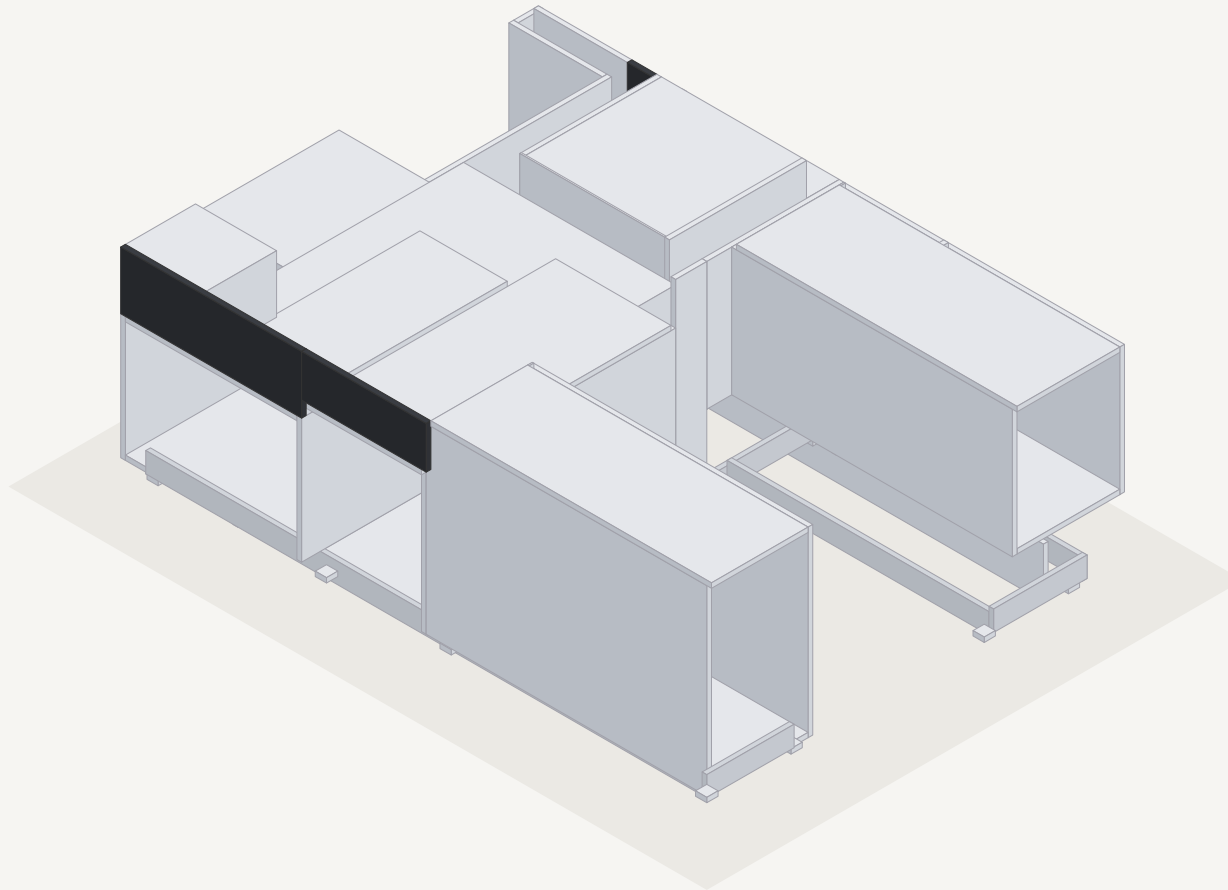


View from the south-west. Parts added in this step are in colour; earlier work is grey.

603-step-03.svg

BUILD STEP 4 - Assemble the base

Carcasses set on the plinth and screwed together; band panels and the utility cover close the faces.

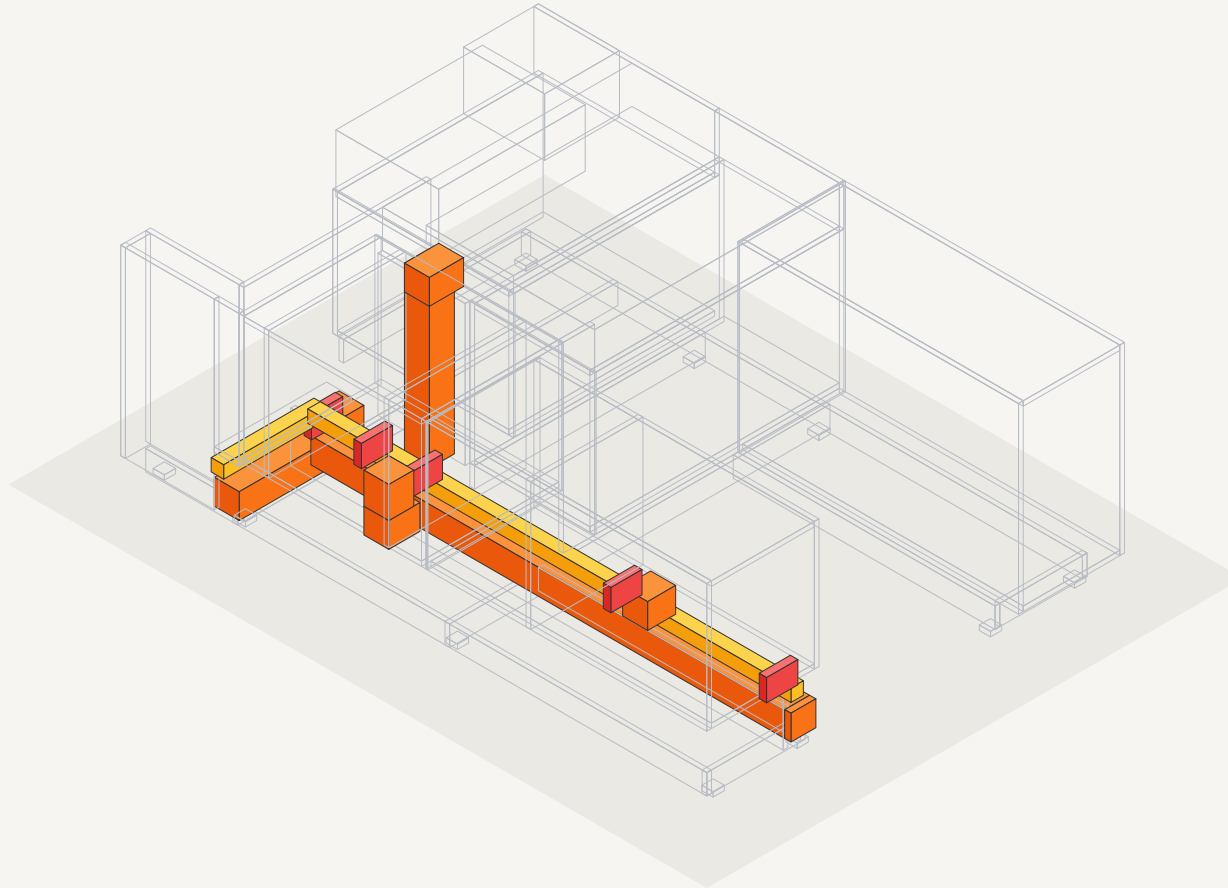


View from the south-west. Parts added in this step are in colour; earlier work is grey.

604-step-04.svg

BUILD STEP 5 - Dust ducting

4" main (orange) and 2-1/2" vac (amber) trunks in the spine, the leg to the east ports, and the five 4" drops with blast gates (red).

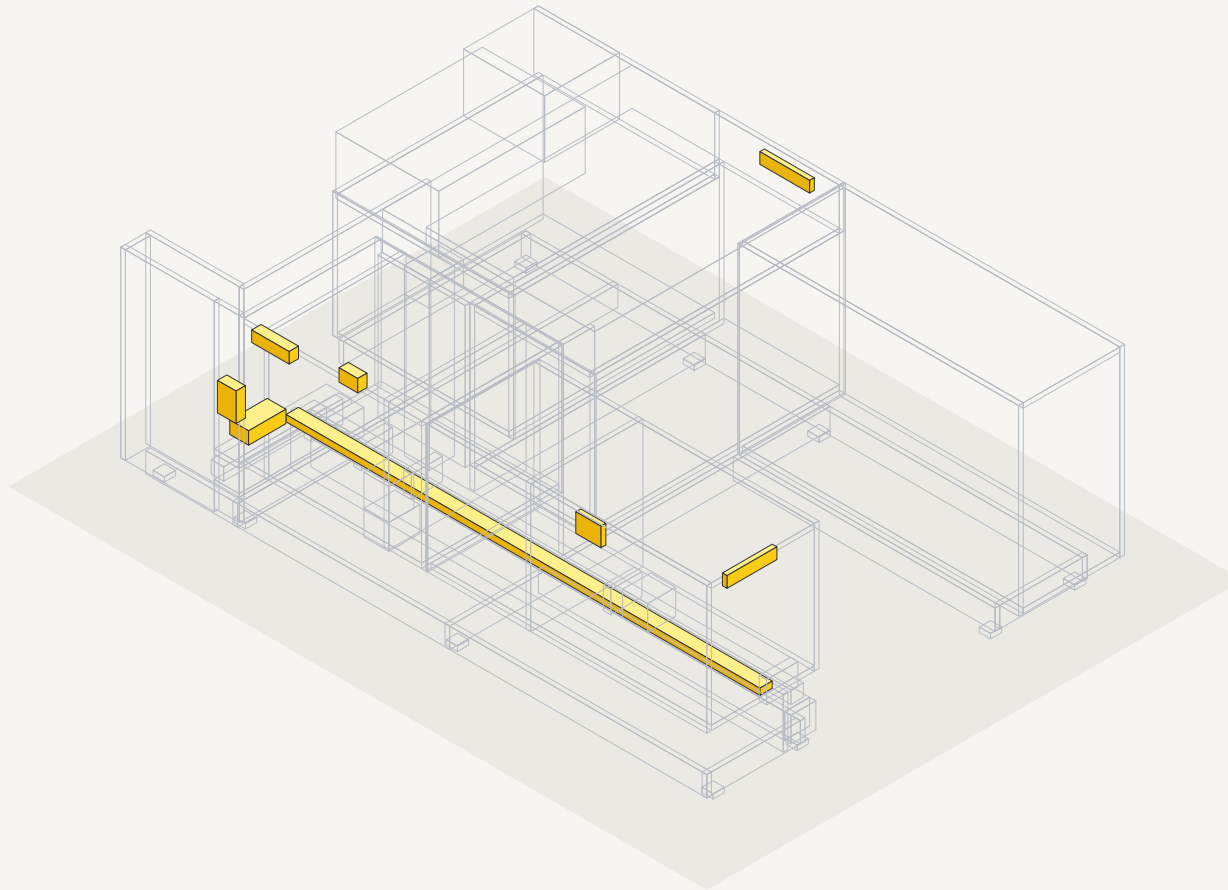


Mirrored view from the north-east so the east face and spine are visible. Parts added in this step are in colour; earlier work is grey wireframe.

605-step-05.svg

BUILD STEP 6 - Electrical

Inlet and main strip in the utility column, raceway along the spine lid, tool outlets in the router box and flip bay (through the seat switch), flush strips on three faces.

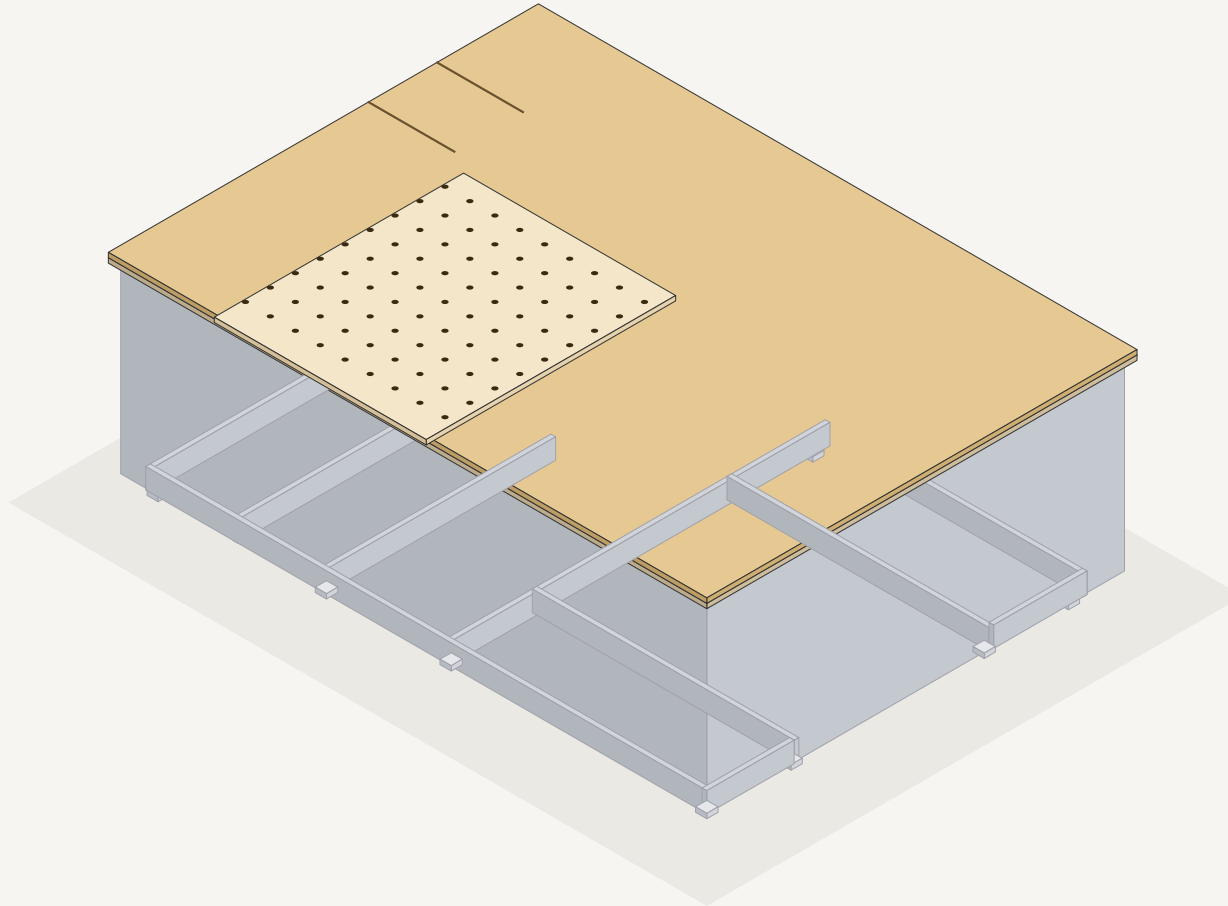


Mirrored view from the north-east so the east face and spine are visible. Parts added in this step are in colour; earlier work is grey wireframe.

606-step-06.svg

BUILD STEP 7 - Laminate and machine the top

Two-layer top glued up (3/4 BB over 3/4 MDF, seams staggered) with the doubler under the dog field; hatch, well, plate and groove cut-outs made.

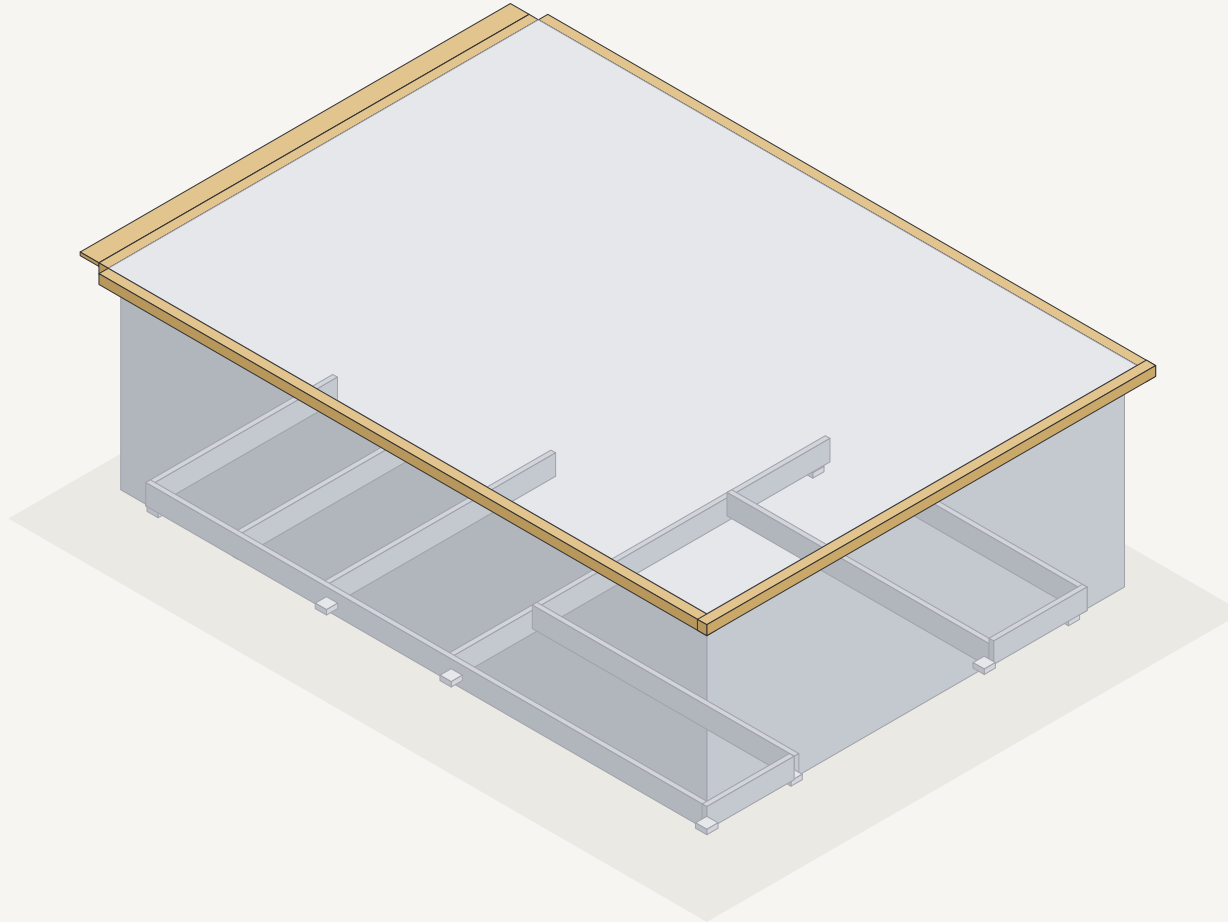


View from the south-west. Parts added in this step are in colour; earlier work is grey.

607-step-07.svg

BUILD STEP 8 - Fit the top, outfeed lip and edging

Top set and screwed from below; maple edging, outfeed beam across the motor bay, bridge lip over the saw's rear rail.

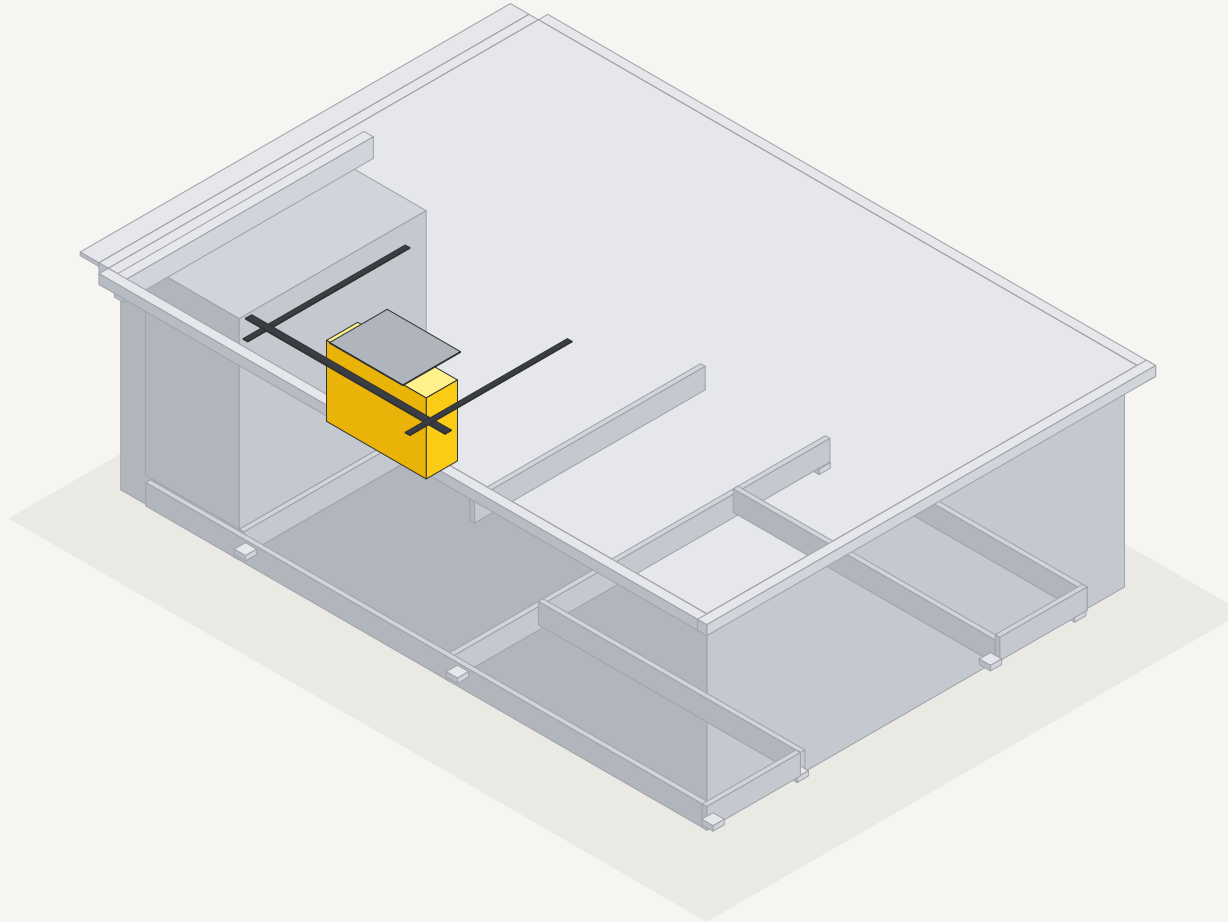


View from the south-west. Parts added in this step are in colour; earlier work is grey.

608-step-08.svg

BUILD STEP 9 - Router station

Router lift plate recess, two fence T-tracks, combo track, router installed in the sealed box, fence vac port.

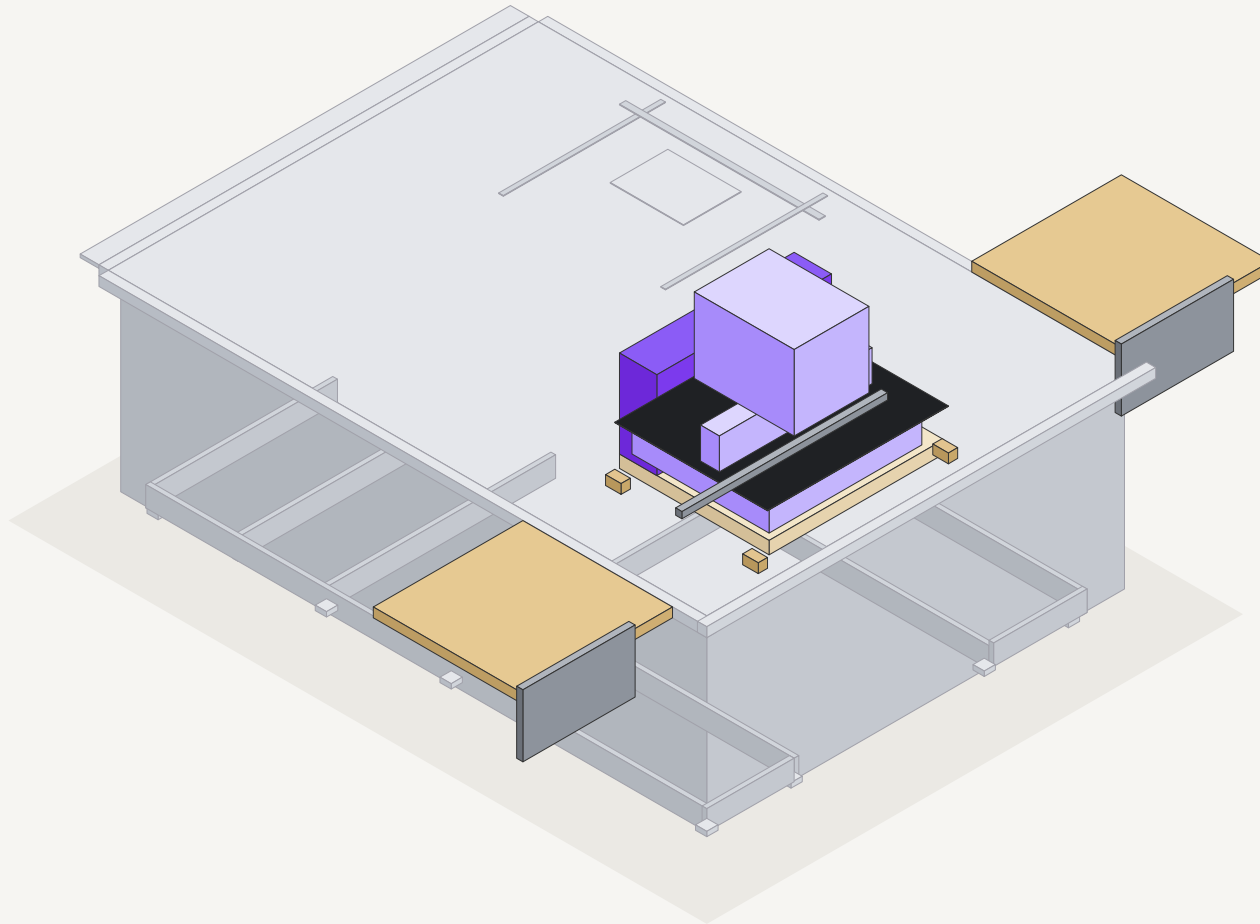


Mirrored view from the north-east so the east face and spine are visible. Parts added in this step are in colour; earlier work is grey.

609-step-09.svg

BUILD STEP 10 - Miter-saw flip-top station

Flip-top platform on its 1" axle: core, flush box, saw and hood; rest blocks, plungers, toggle clamps and the seat switch on the bay walls; drop-leaf wings.

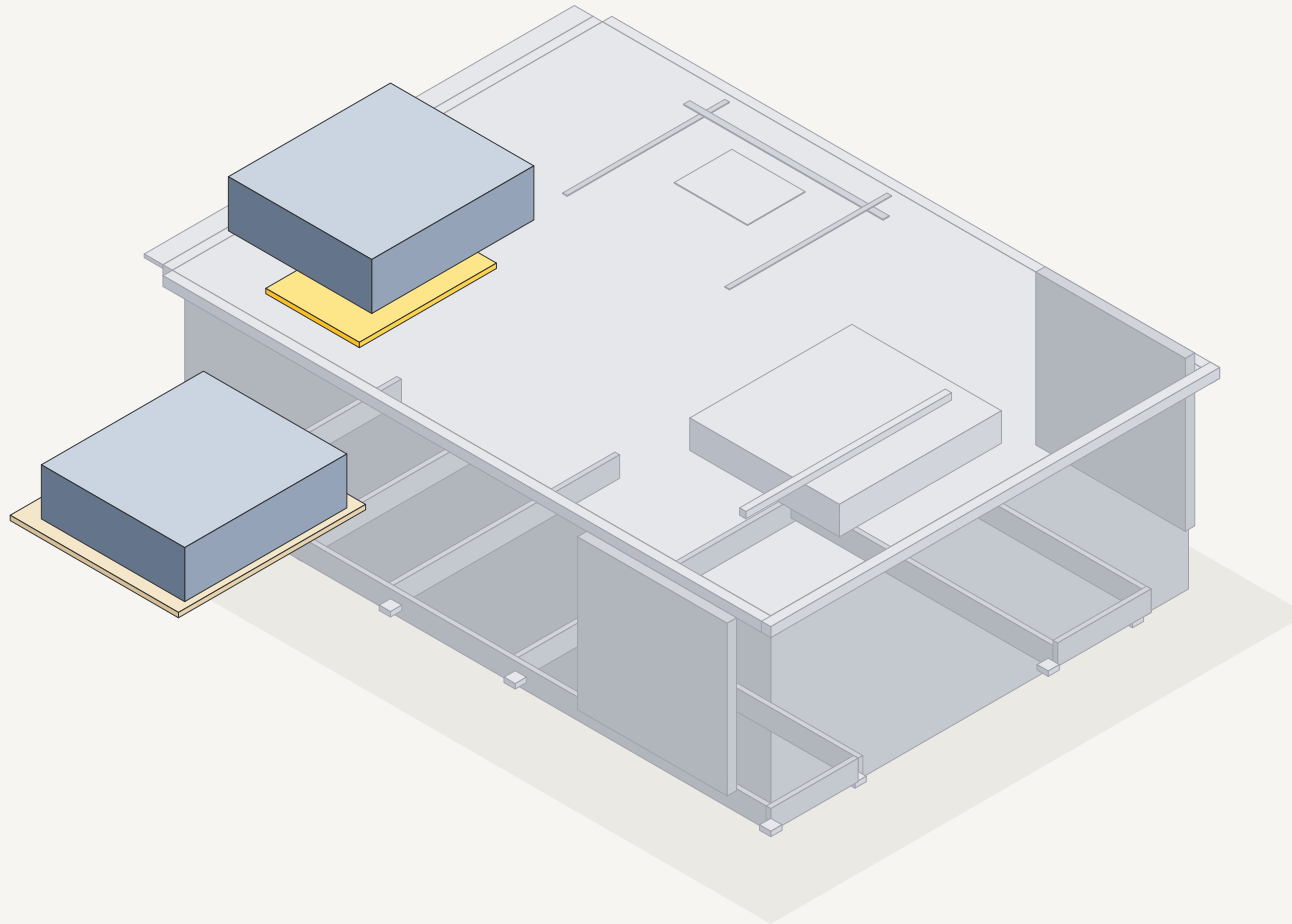


View from the south-west. Parts added in this step are in colour; earlier work is grey.

610-step-10.svg

BUILD STEP 11 - Laser well and tray

Laser well insert and movable floor; laser tray on 30" heavy slides in W2.

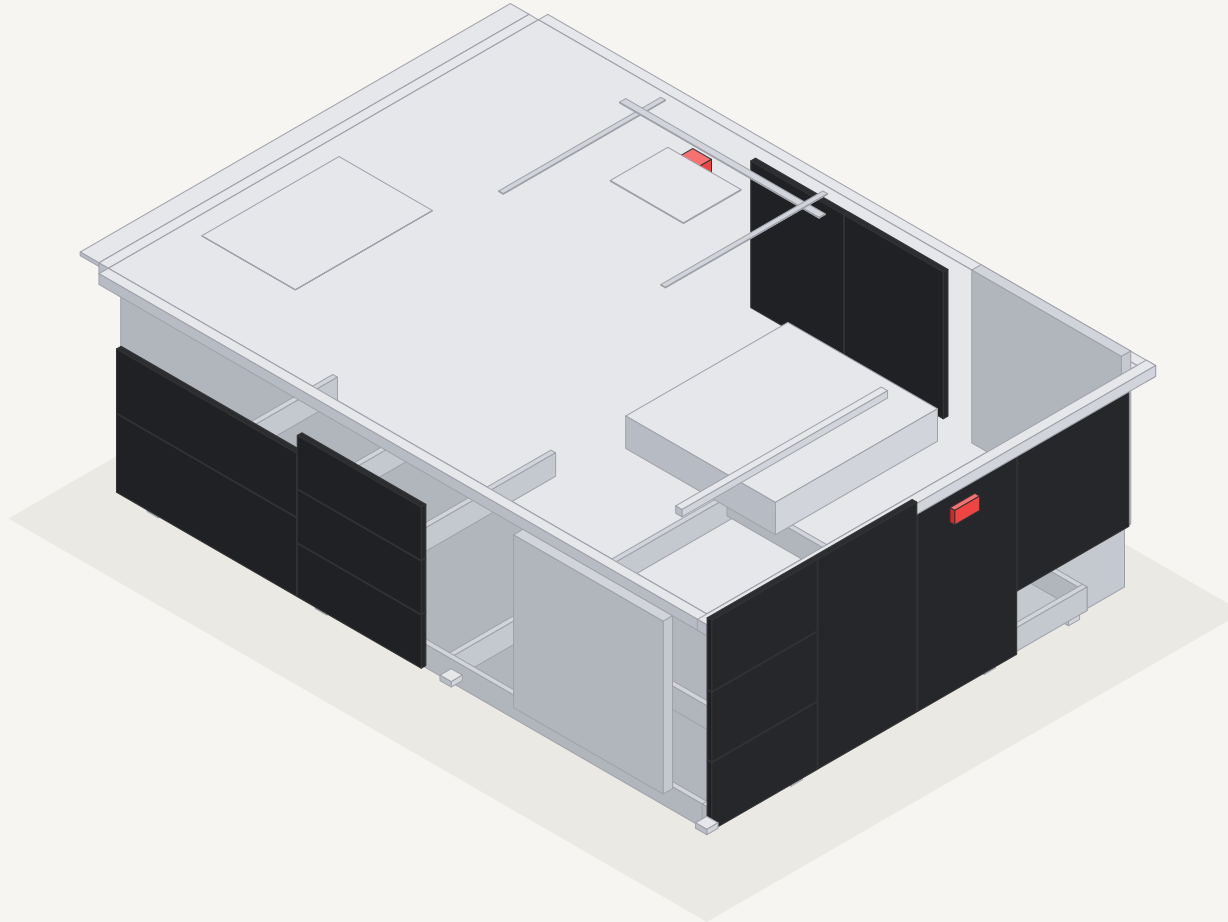


View from the south-west. Parts added in this step are in colour; earlier work is grey.

611-step-11.svg

BUILD STEP 12 - Drawers, pull-outs and fronts

Drawer boxes and all fronts: long drawers, flip-bay doors, clamp rack, bit pull-out, router doors, W1/W2 drawers, laser tray front, paddle switches.

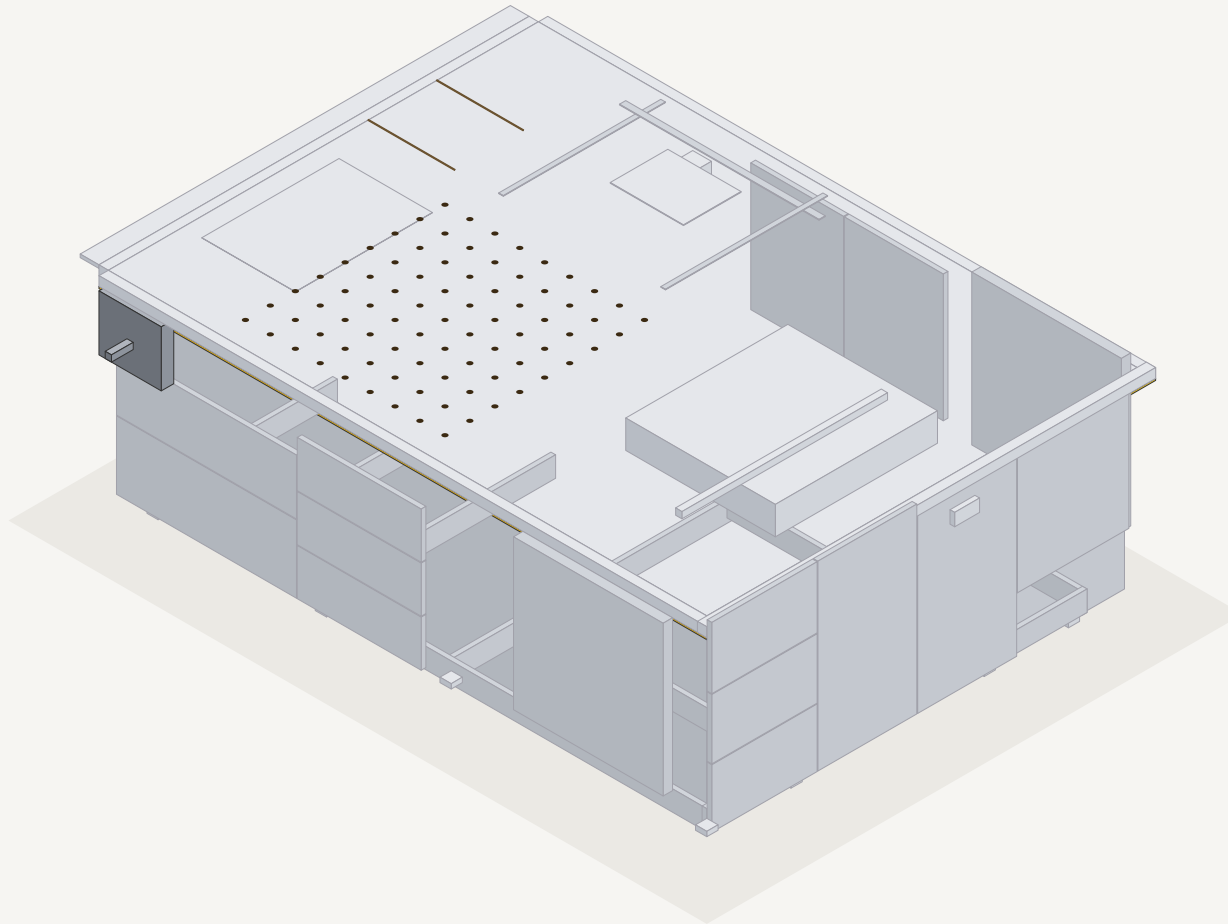


View from the south-west. Parts added in this step are in colour; earlier work is grey.

612-step-12.svg

BUILD STEP 13 - Vise, dogs, lighting and finish

Face vise, dog-hole accessories, pipe-clamp cradles, LED strip and finish.



View from the south-west. Parts added in this step are in colour; earlier work is grey.

613-step-13.svg